

CLASS 10 MATHEMATICS FOUNDATION

Complete Exam-Ready Notes

Structured with Quick Summaries · Real-Life Examples · Formula Sheets

Memory Hooks · Common Mistakes · Chapter-End Revision

Chapter 1: Real Numbers

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Difference between rational and irrational numbers.
- Ability to find HCF and LCM by prime factorisation.
- Knowledge of divisibility and prime numbers.
- Familiarity with decimal expansions.

1.1 Euclid's Division Lemma and Algorithm

 **Estimated time:** 15 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

Euclid's division lemma states that for any integers a and b with $b > 0$ there exist unique q and r with $a = bq + r$ and $0 \leq r < b$, and repeated use of it gives the HCF.

Real-Life Example

Cutting the largest identical tiles that exactly fit two different wall lengths is a HCF problem solved fastest by Euclid's algorithm.

Detailed Explanation

Definition

Euclid's division lemma: given positive integers a and b , there exist unique integers q (quotient) and r (remainder) such that $a = bq + r$, where $0 \leq r < b$. The division algorithm applies this repeatedly to find the HCF.

Key Idea

The remainder condition $0 \leq r < b$ is what makes q and r unique. Without it, infinitely many pairs would satisfy $a = bq + r$.

Working Principle

To find $\text{HCF}(a, b)$ with $a > b$, divide a by b , then divide b by the remainder, and continue. The last non-zero remainder is the HCF.

Applications

- Any positive odd integer can be written as $6q + 1$, $6q + 3$ or $6q + 5$.
- The square of any positive integer is of the form $3m$ or $3m + 1$.
- $\text{HCF}(455, 42) = 7$ by three steps of the algorithm.

Points to Remember (Exam Focus)

- ✓ $a = bq + r$ with $0 \leq r < b$; q and r are unique.
- ✓ Euclid's algorithm: the last non-zero remainder is the HCF.
- ✓ The lemma is used to prove general statements about the form of integers.
- ✓ Every integer is of the form $2q$ or $2q + 1$ (even or odd).
- ✓ The algorithm always terminates because remainders strictly decrease.

⚠ Common Student Mistakes

- ⚠ Allowing $r = b$ or $r > b$; the remainder must be strictly less than the divisor.
- ⚠ Taking the last remainder (zero) as the HCF instead of the last non-zero one.
- ⚠ Applying the lemma with $b = 0$, which is not permitted.

🧠 Memory Hook

'Divide, take the remainder, divide again — the last non-zero remainder is the HCF.'

🚀 Advanced Insights

Euclid's algorithm is one of the oldest algorithms still in daily use and runs in logarithmic time. It underpins modular arithmetic and therefore RSA encryption — ancient mathematics securing modern banking.

Formula Section

📐 Euclid's Division Lemma

Difficulty	● Medium ★★☆☆
Formula	$a = bq + r, 0 \leq r < b$
Meaning of Variables	a = dividend, b = divisor ($b > 0$), q = quotient, r = remainder.
When to Use	To find HCF, or to prove that integers take particular forms.
Common Mistakes	Forgetting the constraint on r , which destroys uniqueness.
Shortcut / Memory Trick	Repeat until the remainder is zero; the previous remainder is the HCF.

1.2 Fundamental Theorem of Arithmetic and Irrationality

🕒 Estimated time: 15 mins Difficulty: ● Medium ★★☆☆ Exam Importance: ★★★★★

🔑 Quick Summary

Every composite number factorises uniquely into primes, and this uniqueness is what proves numbers such as $\sqrt{2}$ to be irrational.

Real-Life Example

The unique prime factorisation of large numbers is exactly what makes RSA encryption secure — factorising is easy to check but hard to do.

Detailed Explanation

Definition

The Fundamental Theorem of Arithmetic states that every composite number can be expressed as a product of primes, and this factorisation is unique apart from the order of the factors.

Key Idea

Uniqueness is the powerful part. It guarantees that if a prime divides a^2 , it must divide a — the single step on which every irrationality proof in this chapter depends.

Working Principle

To prove $\sqrt{2}$ irrational, assume $\sqrt{2} = p/q$ in lowest terms, square to get $p^2 = 2q^2$, deduce that 2 divides p , substitute, deduce that 2 divides q , and contradict the assumption that p/q was in lowest terms.

Applications

- $\text{HCF} \times \text{LCM} = \text{product of the two numbers, for two numbers only.}$
- A rational p/q in lowest terms has a terminating decimal exactly when $q = 2^n 5^m$.
- Proof by contradiction is the standard method for irrationality.

✓ Points to Remember (Exam Focus)

- ✓ Every composite number has a unique prime factorisation.
- ✓ $\text{HCF} = \text{product of the smallest powers of common primes.}$
- ✓ $\text{LCM} = \text{product of the greatest powers of all primes involved.}$
- ✓ $\text{HCF}(a,b) \times \text{LCM}(a,b) = a \times b$ — valid for TWO numbers only.
- ✓ p/q terminates if and only if the denominator in lowest terms is $2^n 5^m$.

Common Student Mistakes

-  Applying $\text{HCF} \times \text{LCM} = \text{product}$ to three numbers, where it fails.
-  Omitting the 'lowest terms' condition when testing for a terminating decimal.
-  Assuming the sum of two irrationals is always irrational.

 Memory Hook

'HCF takes the smallest powers, LCM takes the biggest' — and their product is the product of the numbers.

 Advanced Insights

Unique factorisation fails in some larger number systems. In the ring of numbers $a + b\sqrt{-5}$, the number 6 factorises in two genuinely different ways — which is why the theorem is 'fundamental' rather than obvious.

II Quick Recap — Topics Covered So Far

- Euclid's Division Lemma and Algorithm — Euclid's division lemma states that for any integers a and b with $b > 0$ there exist unique q and r with $a = bq + r$ and $0 \leq r < b$, and repeated use of it gives the HCF.
- Fundamental Theorem of Arithmetic and Irrationality — Every composite number factorises uniquely into primes, and this uniqueness is what proves numbers such as $\sqrt{2}$ to be irrational.

Chapter 1 — End of Chapter Revision

All Memory Hooks

- Euclid's Division Lemma and Algorithm: 'Divide, take the remainder, divide again — the last non-zero remainder is the HCF.'
- Fundamental Theorem of Arithmetic and Irrationality: 'HCF takes the smallest powers, LCM takes the biggest' — and their product is the product of the numbers.

All Common Mistakes

-  Allowing $r = b$ or $r > b$; the remainder must be strictly less than the divisor.
-  Taking the last remainder (zero) as the HCF instead of the last non-zero one.
-  Applying the lemma with $b = 0$, which is not permitted.
-  Applying $\text{HCF} \times \text{LCM} = \text{product}$ to three numbers, where it fails.
-  Omitting the 'lowest terms' condition when testing for a terminating decimal.
-  Assuming the sum of two irrationals is always irrational.

Formula Sheet

Formula	Meaning / Use
Euclid's Division Lemma: $a = bq + r, 0 \leq r < b$	To find HCF, or to prove that integers take particular forms.

Quick Revision Section

- Euclid's Division Lemma and Algorithm: Euclid's division lemma states that for any integers a and b with $b > 0$ there exist unique q and r with $a = bq + r$ and $0 \leq r < b$, and repeated use of it gives the HCF.
- Fundamental Theorem of Arithmetic and Irrationality: Every composite number factorises uniquely into primes, and this uniqueness is what proves numbers such as $\sqrt{2}$ to be irrational.

Chapter 2: Polynomials

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Definition of a polynomial and its degree.
- Factorisation by splitting the middle term.
- Ability to plot a graph from a table of values.
- Knowledge of the remainder and factor theorems.

2.1 Zeroes of a Polynomial and Their Graphical Meaning

 **Estimated time:** 14 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

The zeroes of a polynomial are the x-coordinates of the points where its graph meets the x-axis, and a polynomial of degree n has at most n zeroes.

Real-Life Example

The break-even points of a business, where profit is zero, are exactly the zeroes of the profit polynomial.

Detailed Explanation

Definition

A real number k is a zero of the polynomial $p(x)$ if $p(k) = 0$. Geometrically, the zeroes are the x-coordinates of the points where the graph of $y = p(x)$ intersects the x-axis.

Key Idea

The graph tells you the number of real zeroes at a glance. A parabola cutting the axis twice has two zeroes, touching it once has one repeated zero, and missing it entirely has none.

Working Principle

A linear polynomial has exactly one zero, a quadratic at most two and a cubic at most three. The graph of a quadratic is a parabola, opening upward if $a > 0$ and downward if $a < 0$.

Applications

- $y = x^2 - 4$ cuts the axis at $x = 2$ and $x = -2$.
- $y = x^2 + 1$ never meets the x-axis, so it has no real zeroes.
- $y = (x - 3)^2$ touches the axis at $x = 3$, a repeated zero.

Points to Remember (Exam Focus)

- ✓ A polynomial of degree n has AT MOST n real zeroes.
- ✓ Zeroes are the x -coordinates of the graph's intersections with the x -axis.
- ✓ A quadratic graph is a parabola: upward if $a > 0$, downward if $a < 0$.
- ✓ Touching the x -axis means a repeated (equal) zero.
- ✓ A cubic can have one, two or three real zeroes.

⚠ Common Student Mistakes

- ⚠ Reading a y -intercept as a zero; zeroes lie on the x -axis.
- ⚠ Assuming every quadratic has two real zeroes.
- ⚠ Saying a degree- n polynomial has exactly n real zeroes; it has at most n .

🧠 Memory Hook

'Zeroes are where the curve crosses the x -axis' — count the crossings, count the zeroes.

🚀 Advanced Insights

The discriminant $b^2 - 4ac$ decides the number of crossings before the graph is drawn: positive gives two, zero gives one, negative gives none. Algebra predicts the geometry exactly.

2.2 Relationship Between Zeroes and Coefficients

🕒 **Estimated time:** 15 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

🔑 Quick Summary

For a quadratic $ax^2 + bx + c$, the sum of the zeroes is $-b/a$ and their product is c/a , which lets a polynomial be reconstructed from its zeroes.

🌍 Real-Life Example

Knowing only the total and the product of two unknown quantities is enough to write down the quadratic they satisfy — a standard trick in word problems.

📖 Detailed Explanation

Definition

If α and β are the zeroes of $ax^2 + bx + c$, then $\alpha + \beta = -b/a$ and $\alpha\beta = c/a$. For a cubic $ax^3 + bx^2 + cx + d$ with zeroes α, β, γ : $\alpha + \beta + \gamma = -b/a$, $\alpha\beta + \beta\gamma + \gamma\alpha = c/a$, $\alpha\beta\gamma = -d/a$.

Key Idea

The signs alternate as the degree of the symmetric expression increases: $-b/a$, $+c/a$, $-d/a$. Remembering the alternating pattern is easier than memorising each result.

Working Principle

A quadratic with given zeroes is written as $k[x^2 - (\text{sum})x + (\text{product})]$, where k is any non-zero constant. Taking $k = 1$ gives the simplest such polynomial.

Applications

- Zeroes 3 and -2 give $x^2 - x - 6$.
- Given one zero of a cubic, the other two are found by dividing and solving the quotient quadratic.
- Symmetric expressions such as $\alpha^2 + \beta^2$ are evaluated as $(\alpha + \beta)^2 - 2\alpha\beta$.

✓ Points to Remember (Exam Focus)

- ✓ Quadratic: $\alpha + \beta = -b/a$, $\alpha\beta = c/a$.
- ✓ Cubic: $\Sigma\alpha = -b/a$, $\Sigma\alpha\beta = c/a$, $\alpha\beta\gamma = -d/a$.
- ✓ Polynomial from zeroes: $k[x^2 - (\alpha + \beta)x + \alpha\beta]$.
- ✓ $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$.
- ✓ $1/\alpha + 1/\beta = (\alpha + \beta)/\alpha\beta$.

⚠ Common Student Mistakes

- ⚠ Writing the sum of the zeroes as $+b/a$ instead of $-b/a$.
- ⚠ Forgetting to divide by a when the leading coefficient is not 1.
- ⚠ Omitting the arbitrary constant k when constructing a polynomial from its zeroes.

🧠 Memory Hook

'Sum is minus b over a, Product is c over a' — and the signs alternate as you go up in degree.

🚀 Advanced Insights

These are Vieta's formulas, valid for polynomials of any degree. They connect the coefficients to the elementary symmetric functions of the roots, a link that opens into Galois theory.

Formula Section**📐 Zeroes and Coefficients of a Quadratic**

Difficulty	● Medium ★★☆☆
Formula	$\alpha + \beta = -b/a$, $\alpha\beta = c/a$

Meaning of Variables	a, b, c = coefficients of $ax^2 + bx + c$; α, β = the two zeroes.
When to Use	To find the sum or product of zeroes without solving, or to build a polynomial from its zeroes.
Common Mistakes	Dropping the minus sign in the sum.
Shortcut / Memory Trick	Quadratic with given zeroes: $x^2 - (\text{sum})x + (\text{product})$.

II Quick Recap — Topics Covered So Far

- **Zeroes of a Polynomial and Their Graphical Meaning** — The zeroes of a polynomial are the x-coordinates of the points where its graph meets the x-axis, and a polynomial of degree n has at most n zeroes.
- **Relationship Between Zeroes and Coefficients** — For a quadratic $ax^2 + bx + c$, the sum of the zeroes is $-b/a$ and their product is c/a , which lets a polynomial be reconstructed from its zeroes.

Chapter 2 — End of Chapter Revision

All Memory Hooks

- Zeroes of a Polynomial and Their Graphical Meaning: 'Zeroes are where the curve crosses the x-axis' — count the crossings, count the zeroes.
- Relationship Between Zeroes and Coefficients: 'Sum is minus b over a, Product is c over a' — and the signs alternate as you go up in degree.

All Common Mistakes

-  Reading a y-intercept as a zero; zeroes lie on the x-axis.
-  Assuming every quadratic has two real zeroes.
-  Saying a degree-n polynomial has exactly n real zeroes; it has at most n.
-  Writing the sum of the zeroes as $+b/a$ instead of $-b/a$.
-  Forgetting to divide by a when the leading coefficient is not 1.
-  Omitting the arbitrary constant k when constructing a polynomial from its zeroes.

Formula Sheet

Formula	Meaning / Use
Zeroes and Coefficients of a Quadratic: $\alpha + \beta = -b/a$, $\alpha\beta = c/a$	To find the sum or product of zeroes without solving, or to build a polynomial from its zeroes.

Quick Revision Section

- Zeroes of a Polynomial and Their Graphical Meaning: The zeroes of a polynomial are the x-coordinates of the points where its graph meets the x-axis, and a polynomial of degree n has at most n zeroes.
- Relationship Between Zeroes and Coefficients: For a quadratic $ax^2 + bx + c$, the sum of the zeroes is $-b/a$ and their product is c/a , which lets a polynomial be reconstructed from its zeroes.

Chapter 3: Pair of Linear Equations in Two Variables

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Ability to solve a linear equation in one variable.
- Plotting a straight line from its equation.
- Understanding of ratios and proportion.
- Familiarity with the Cartesian plane.

3.1 Graphical Method and Consistency

 **Estimated time:** 15 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

Two linear equations represent two lines, which may intersect at one point, be parallel, or coincide — giving one solution, none, or infinitely many.

Real-Life Example

Comparing two mobile tariff plans graphically shows immediately whether one is always cheaper or whether they cross at some usage level.

Detailed Explanation

Definition

A pair of linear equations $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ is consistent if it has at least one solution and inconsistent if it has none. A consistent pair with infinitely many solutions is called dependent.

Key Idea

The ratios of the coefficients decide everything without any drawing. Comparing a_1/a_2 , b_1/b_2 and c_1/c_2 tells you the number of solutions in a single line of work.

Working Principle

If $a_1/a_2 \neq b_1/b_2$, the lines intersect and there is a unique solution. If all three ratios are equal, the lines coincide. If the first two are equal but differ from the third, the lines are parallel.

Applications

- Intersecting lines: unique solution, consistent and independent.
- Coincident lines: infinitely many solutions, consistent and dependent.
- Parallel lines: no solution, inconsistent.

Points to Remember (Exam Focus)

- ✓ $a_1/a_2 \neq b_1/b_2 \rightarrow$ unique solution (intersecting lines).
- ✓ $a_1/a_2 = b_1/b_2 = c_1/c_2 \rightarrow$ infinitely many solutions (coincident lines).
- ✓ $a_1/a_2 = b_1/b_2 \neq c_1/c_2 \rightarrow$ no solution (parallel lines).
- ✓ Consistent means at least one solution exists.
- ✓ Both equations must be in the form $ax + by + c = 0$ before comparing ratios.

⚠ Common Student Mistakes

- ⚠ Comparing ratios before writing both equations in standard form.
- ⚠ Calling a dependent pair inconsistent; it is consistent with infinitely many solutions.
- ⚠ Mixing up the parallel and coincident conditions, which differ only in the third ratio.

🧠 Memory Hook

'Two ratios differ $\rightarrow$ one point; all three equal $\rightarrow$ same line; only the last differs $\rightarrow$ never meet.'

🚀 Advanced Insights

In matrix terms the pair has a unique solution exactly when the coefficient determinant $a_1b_2 - a_2b_1$ is non-zero. Cramer's rule then gives the solution directly, and the ratio conditions are that determinant in disguise.

3.2 Algebraic Methods of Solution

🕒 Estimated time: 15 mins Difficulty: ● Medium ★★☆☆ Exam Importance: ★★★★★

🔍 Quick Summary

A pair of linear equations is solved algebraically by substitution, elimination or cross-multiplication, whichever suits the coefficients best.

🌐 Real-Life Example

Splitting a bill where two items have unknown prices but two different totals are known is solved in seconds by elimination.

📖 Detailed Explanation

Definition

In substitution, one variable is expressed in terms of the other and substituted. In elimination, the equations are added or subtracted after equalising one coefficient. Cross-multiplication uses a direct formula.

Key Idea

Elimination is usually fastest when a coefficient is already equal or easily made so; substitution is easier when one variable has coefficient 1. Choosing well saves most of the arithmetic.

Working Principle

Equations reducible to linear form, such as those involving $1/x$ and $1/y$, are solved by substituting $u = 1/x$ and $v = 1/y$, solving the linear pair, and converting back.

Applications

- Age problems, speed–distance–time problems, and work problems.
- Boat and stream problems: upstream speed = $b - s$, downstream = $b + s$.
- Fraction problems where numerator and denominator are both unknown.

✓ Points to Remember (Exam Focus)

- ✓ Substitution: express one variable and substitute into the other equation.
- ✓ Elimination: make one coefficient equal, then add or subtract.
- ✓ Cross-multiplication: $x/(b_1c_2 - b_2c_1) = y/(c_1a_2 - c_2a_1) = 1/(a_1b_2 - a_2b_1)$.
- ✓ Equations in $1/x$ and $1/y$ become linear after substituting $u = 1/x$, $v = 1/y$.
- ✓ Always verify the answer in BOTH original equations.

⚠ Common Student Mistakes

- ⚠ Forgetting to convert u and v back to x and y at the end.
- ⚠ Adding when the signs require subtraction during elimination.
- ⚠ Checking the solution in only one of the two equations.

🧠 Memory Hook

'Coefficient of 1 → substitute; matching coefficients → eliminate.'

🚀 Advanced Insights

Cross-multiplication is Cramer's rule written out. For larger systems, Gaussian elimination generalises the elimination method and is what computers actually use to solve thousands of equations at once.

Formula Section**📐 Cross-Multiplication Method**

Difficulty	● Hard ★★★
Formula	$x/(b_1c_2 - b_2c_1) = y/(c_1a_2 - c_2a_1) = 1/(a_1b_2 - a_2b_1)$

Meaning of Variables	a_1, b_1, c_1 and a_2, b_2, c_2 = coefficients of the two equations in the form $ax + by + c = 0$.
When to Use	When both equations are already in standard form and the coefficients are awkward.
Common Mistakes	Not writing both equations with all terms on the left before applying the formula.
Shortcut / Memory Trick	If $a_1b_2 - a_2b_1 = 0$, there is no unique solution — check the ratios instead.

II Quick Recap — Topics Covered So Far

- Graphical Method and Consistency — Two linear equations represent two lines, which may intersect at one point, be parallel, or coincide — giving one solution, none, or infinitely many.
- Algebraic Methods of Solution — A pair of linear equations is solved algebraically by substitution, elimination or cross-multiplication, whichever suits the coefficients best.

Chapter 3 — End of Chapter Revision

All Memory Hooks

- Graphical Method and Consistency: 'Two ratios differ → one point; all three equal → same line; only the last differs → never meet.'
- Algebraic Methods of Solution: 'Coefficient of 1 → substitute; matching coefficients → eliminate.'

All Common Mistakes

-  Comparing ratios before writing both equations in standard form.
-  Calling a dependent pair inconsistent; it is consistent with infinitely many solutions.
-  Mixing up the parallel and coincident conditions, which differ only in the third ratio.
-  Forgetting to convert u and v back to x and y at the end.
-  Adding when the signs require subtraction during elimination.
-  Checking the solution in only one of the two equations.

Formula Sheet

Formula	Meaning / Use
Cross-Multiplication Method: $x/(b_1c_2 - b_2c_1) = y/(c_1a_2 - c_2a_1) = 1/(a_1b_2 - a_2b_1)$	When both equations are already in standard form and the coefficients are awkward.

Quick Revision Section

- Graphical Method and Consistency: Two linear equations represent two lines, which may intersect at one point, be parallel, or coincide — giving one solution, none, or infinitely many.
- Algebraic Methods of Solution: A pair of linear equations is solved algebraically by substitution, elimination or cross-multiplication, whichever suits the coefficients best.

Chapter 4: Quadratic Equations

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Factorisation by splitting the middle term.
- Ability to simplify surds and square roots.
- Relationship between zeroes and coefficients of a polynomial.
- Skill in framing equations from word problems.

4.1 Solving Quadratic Equations

 **Estimated time:** 16 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

A quadratic equation $ax^2 + bx + c = 0$ with $a \neq 0$ is solved by factorisation, by completing the square, or by the quadratic formula.

Real-Life Example

Finding the width of a uniform path around a rectangular garden of given total area leads directly to a quadratic equation.

Detailed Explanation

Definition

A quadratic equation is an equation of the form $ax^2 + bx + c = 0$ where a , b and c are real numbers and $a \neq 0$. A root is a value of x satisfying the equation.

Key Idea

Factorisation is fastest when it works, but it works only for nice coefficients. The quadratic formula always works, which is why it is derived by completing the square once and for all.

Working Principle

Factorisation uses the zero-product rule: if $pq = 0$ then $p = 0$ or $q = 0$. Completing the square rewrites the equation as a perfect square equal to a constant.

Applications

- Splitting the middle term: find two numbers with product ac and sum b .
- Word problems on speed, age, area and numbers commonly reduce to quadratics.
- Always reject roots that are physically impossible, such as a negative length.

Points to Remember (Exam Focus)

- ✓ Standard form $ax^2 + bx + c = 0$ with $a \neq 0$.
- ✓ A quadratic equation has at most two roots.
- ✓ Zero-product rule: if $pq = 0$ then $p = 0$ or $q = 0$.
- ✓ The quadratic formula works for every quadratic; factorisation does not.
- ✓ In word problems, discard roots that make no physical sense.

⚠ Common Student Mistakes

- ⚠ Applying the zero-product rule when the product is not zero.
- ⚠ Forgetting to bring all terms to one side before solving.
- ⚠ Keeping a negative root for a length, age or number of items.

🧠 Memory Hook

'All terms to one side, then factorise or use the formula.'

🚀 Advanced Insights

The quadratic formula is simply completing the square carried out with letters instead of numbers. Doing it once symbolically saves repeating the process for every individual equation.

Formula Section

📐 Quadratic Formula

Difficulty	● Medium ★★☆☆
Formula	$x = [-b \pm \sqrt{b^2 - 4ac}] / 2a$
Meaning of Variables	a, b, c = coefficients of $ax^2 + bx + c = 0$.
When to Use	For any quadratic, especially when factorisation is not obvious.
Common Mistakes	Dividing only part of the numerator by $2a$, or dropping the $\pm$ sign.
Shortcut / Memory Trick	Compute the discriminant first — if it is negative, stop, there are no real roots.

4.2 Nature of Roots and the Discriminant

🕒 Estimated time: 13 mins Difficulty: ● Medium ★★☆☆ Exam Importance: ★★★★★

 Quick Summary

The discriminant $D = b^2 - 4ac$ decides the nature of the roots without solving the equation at all.

 Real-Life Example

An engineer checking whether a projectile will ever reach a given height only needs the sign of the discriminant, not the actual times.

 Detailed Explanation**Definition**

The discriminant of $ax^2 + bx + c = 0$ is $D = b^2 - 4ac$. It is the quantity under the square root in the quadratic formula.

Key Idea

The sign of D alone answers the question of how many real roots exist. Positive gives two distinct real roots, zero gives two equal roots, and negative gives no real roots.

Working Principle

Questions asking for the value of k such that an equation has equal roots are solved by setting $D = 0$ and solving the resulting equation in k .

Applications

- Graphically, $D > 0$ means the parabola cuts the x -axis twice, $D = 0$ once, $D < 0$ not at all.
- The roots are rational when D is a perfect square and a, b, c are rational.
- Sum of roots $= -b/a$ and product $= c/a$, whatever the sign of D .

 Points to Remember (Exam Focus)

- ✓ $D = b^2 - 4ac$.
- ✓ $D > 0$: two distinct real roots. $D = 0$: two equal real roots. $D < 0$: no real roots.
- ✓ For equal roots, set $D = 0$ and solve for the unknown parameter.
- ✓ Roots are rational if D is a perfect square and the coefficients are rational.
- ✓ Sum of roots $= -b/a$; product $= c/a$.

 Common Student Mistakes

-  Writing the discriminant as $b^2 + 4ac$.
-  Saying an equation with $D < 0$ has no roots at all; it has no REAL roots.
-  Forgetting to check that $a \neq 0$ before treating an equation as quadratic.

 Memory Hook

'D positive, two roots; D zero, one repeated; D negative, none at all.'

 Advanced Insights

When $D < 0$ the roots are complex conjugates. Extending to complex numbers makes the count of roots exactly two for every quadratic, which is the Fundamental Theorem of Algebra in its simplest case.

Formula Section **Discriminant**

Difficulty	● Easy ★☆☆
Formula	$D = b^2 - 4ac$
Meaning of Variables	a, b, c = coefficients of the quadratic equation.
When to Use	To determine the nature of the roots, or to find a parameter giving equal roots.
Common Mistakes	Using + instead of – in the formula.
Shortcut / Memory Trick	Equal roots means $D = 0$ — this is the standard route for 'find the value of k ' questions.

II Quick Recap — Topics Covered So Far

- Solving Quadratic Equations — A quadratic equation $ax^2 + bx + c = 0$ with $a \neq 0$ is solved by factorisation, by completing the square, or by the quadratic formula.
- Nature of Roots and the Discriminant — The discriminant $D = b^2 - 4ac$ decides the nature of the roots without solving the equation at all.

Chapter 4 — End of Chapter Revision

All Memory Hooks

- Solving Quadratic Equations: 'All terms to one side, then factorise or use the formula.'
- Nature of Roots and the Discriminant: 'D positive, two roots; D zero, one repeated; D negative, none at all.'

All Common Mistakes

-  Applying the zero-product rule when the product is not zero.
-  Forgetting to bring all terms to one side before solving.
-  Keeping a negative root for a length, age or number of items.
-  Writing the discriminant as $b^2 + 4ac$.
-  Saying an equation with $D < 0$ has no roots at all; it has no REAL roots.
-  Forgetting to check that $a \neq 0$ before treating an equation as quadratic.

Formula Sheet

Formula	Meaning / Use
Quadratic Formula: $x = [-b \pm \sqrt{b^2 - 4ac}] / 2a$	For any quadratic, especially when factorisation is not obvious.
Discriminant: $D = b^2 - 4ac$	To determine the nature of the roots, or to find a parameter giving equal roots.

Quick Revision Section

- Solving Quadratic Equations: A quadratic equation $ax^2 + bx + c = 0$ with $a \neq 0$ is solved by factorisation, by completing the square, or by the quadratic formula.
- Nature of Roots and the Discriminant: The discriminant $D = b^2 - 4ac$ decides the nature of the roots without solving the equation at all.

Chapter 5: Arithmetic Progression

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Ability to identify patterns in a sequence of numbers.
- Comfort with substituting into a formula.
- Skill in solving linear and quadratic equations.
- Understanding of the term 'common difference'.

5.1 Arithmetic Progression and the n th Term

 **Estimated time:** 15 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

An arithmetic progression is a sequence in which each term differs from the previous one by a fixed common difference d , and its n th term is $a + (n - 1)d$.

Real-Life Example

A salary rising by a fixed increment each year, or the seats increasing row by row in an auditorium, are arithmetic progressions.

Detailed Explanation

Definition

An arithmetic progression (AP) is a list of numbers in which each term, except the first, is obtained by adding a fixed number d , the common difference, to the preceding term.

Key Idea

The coefficient of d is $(n - 1)$, not n , because the first term needs no addition at all. This off-by-one is the most frequent source of error in the whole chapter.

Working Principle

To check whether a sequence is an AP, verify that $a_2 - a_1 = a_3 - a_2 = a_4 - a_3$. A single equal difference is not enough.

Applications

- The n th term from the end of an AP with l as the last term is $l - (n - 1)d$.
- Three terms in AP are conveniently taken as $a - d$, a , $a + d$.
- A negative d gives a decreasing AP.

Points to Remember (Exam Focus)

- ✓ $a_n = a + (n - 1)d$, where a is the first term and d the common difference.
- ✓ $d = a_n - a_{n-1}$, and must be the same throughout.
- ✓ The n th term from the end $= l - (n - 1)d$.
- ✓ For three terms in AP, take $a - d$, a , $a + d$ to simplify algebra.
- ✓ $d > 0$ gives an increasing AP; $d < 0$ a decreasing one.

⚠ Common Student Mistakes

- ⚠ Using $a + nd$ instead of $a + (n - 1)d$.
- ⚠ Checking only one pair of consecutive terms before declaring a sequence an AP.
- ⚠ Assuming d must be positive.



Memory Hook

'First term needs no step, so multiply d by $(n - 1)$, never n .'



Advanced Insights

An AP is a linear function of n , so its graph is a set of collinear points. This is why the sum formula is quadratic in n — summing a linear function always gives a quadratic.

Formula Section



n th Term of an AP

Difficulty	● Easy ★☆☆
Formula	$a_n = a + (n - 1)d$
Meaning of Variables	a = first term, d = common difference, n = term number.
When to Use	To find any term, or to find n when a particular term is given.
Common Mistakes	Using n instead of $(n - 1)$ as the multiplier of d .
Shortcut / Memory Trick	If n comes out as a non-integer, that value is not a term of the AP.

5.2 Sum of n Terms of an AP

🕒 Estimated time: 15 mins Difficulty: ● Medium ★★☆☆ Exam Importance: ★★★★★



Quick Summary

The sum of the first n terms of an AP is $S_n = (n/2)[2a + (n - 1)d]$, which equals $(n/2)(a + l)$ when the last term is known.

Real-Life Example

Total savings over ten years with a fixed annual increase is the sum of an AP — no need to add the ten amounts one by one.

Detailed Explanation

Definition

S_n denotes the sum of the first n terms of an arithmetic progression. It is given by $S_n = (n/2)[2a + (n - 1)d]$ or equivalently $(n/2)(a + l)$, where l is the last term.

Key Idea

The second form is quicker whenever the last term is known, because it avoids computing d altogether. Choosing the right form halves the work.

Working Principle

The formula comes from pairing the first and last terms, the second and second-last, and so on. Each pair sums to $a + l$, and there are $n/2$ such pairs.

Applications

- $a_n = S_n - S_{n-1}$ recovers a term from the sums.
- Sum of the first n natural numbers = $n(n+1)/2$.
- Sum of the first n odd numbers = n^2 .

✓ Points to Remember (Exam Focus)

- ✓ $S_n = (n/2)[2a + (n - 1)d]$.
- ✓ $S_n = (n/2)(a + l)$ when the last term l is known.
- ✓ $a_n = S_n - S_{n-1}$.
- ✓ Sum of the first n natural numbers = $n(n+1)/2$.
- ✓ If S_n is quadratic in n , the sequence is an AP.

Common Student Mistakes

-  Using $(n/2)(a + l)$ when l is not the n th term.
-  Forgetting the factor $n/2$ outside the bracket.
-  Solving a quadratic in n and keeping a negative or fractional value.

 Memory Hook

'Half the number of terms, times first plus last' — Gauss's schoolboy trick.

 Advanced Insights

Gauss reportedly summed 1 to 100 in seconds by pairing 1+100, 2+99, and so on. The general formula is nothing more than that pairing written in symbols.

Formula Section

 Sum of n Terms of an AP

Difficulty	● Medium ★★☆☆
Formula	$S_n = (n/2)[2a + (n - 1)d] = (n/2)(a + l)$
Meaning of Variables	n = number of terms, a = first term, d = common difference, l = last term.
When to Use	To find a total, or to find n given the sum.
Common Mistakes	Mixing the two forms and using both d and l in the same expression.
Shortcut / Memory Trick	Use $(n/2)(a + l)$ whenever the last term is given.

II Quick Recap — Topics Covered So Far

- Arithmetic Progression and the nth Term — An arithmetic progression is a sequence in which each term differs from the previous one by a fixed common difference d , and its n th term is $a + (n - 1)d$.
- Sum of n Terms of an AP — The sum of the first n terms of an AP is $S_n = (n/2)[2a + (n - 1)d]$, which equals $(n/2)(a + l)$ when the last term is known.

Chapter 5 — End of Chapter Revision

All Memory Hooks

- Arithmetic Progression and the n th Term: 'First term needs no step, so multiply d by $(n - 1)$, never n .'
- Sum of n Terms of an AP: 'Half the number of terms, times first plus last' — Gauss's schoolboy trick.

All Common Mistakes

-  Using $a + nd$ instead of $a + (n - 1)d$.
-  Checking only one pair of consecutive terms before declaring a sequence an AP.
-  Assuming d must be positive.
-  Using $(n/2)(a + l)$ when l is not the n th term.
-  Forgetting the factor $n/2$ outside the bracket.
-  Solving a quadratic in n and keeping a negative or fractional value.

Formula Sheet

Formula	Meaning / Use
nth Term of an AP: $a_n = a + (n - 1)d$	To find any term, or to find n when a particular term is given.
Sum of n Terms of an AP: $S_n = (n/2)[2a + (n - 1)d] = (n/2)(a + l)$	To find a total, or to find n given the sum.

Quick Revision Section

- Arithmetic Progression and the n th Term: An arithmetic progression is a sequence in which each term differs from the previous one by a fixed common difference d , and its n th term is $a + (n - 1)d$.
- Sum of n Terms of an AP: The sum of the first n terms of an AP is $S_n = (n/2)[2a + (n - 1)d]$, which equals $(n/2)(a + l)$ when the last term is known.

Chapter 6: Triangles

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Congruence criteria for triangles.
- Properties of parallel lines and a transversal.
- Ratio and proportion.
- Area of a triangle as $\frac{1}{2} \times \text{base} \times \text{height}$.

6.1 Similar Triangles and Criteria for Similarity

 **Estimated time:** 16 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

Two triangles are similar if their corresponding angles are equal and corresponding sides are proportional, established by the AAA, SSS or SAS similarity criteria.

Real-Life Example

The height of a tree is found by comparing its shadow with that of a metre stick — two similar triangles doing the measurement for you.

Detailed Explanation

Definition

Two triangles are similar if their corresponding angles are equal and their corresponding sides are in the same ratio. Congruent triangles are a special case of similar triangles with ratio 1.

Key Idea

For triangles, equal angles automatically force proportional sides and vice versa. This is peculiar to triangles — for quadrilaterals both conditions must be checked separately.

Working Principle

The AA criterion is enough because the third angle is then determined by the angle sum property. In practice AA is the criterion used in almost every problem.

Applications

- Basic Proportionality Theorem (Thales): a line parallel to one side divides the other two proportionally.
- Its converse is used to prove that a line is parallel to a side.
- The ratio of areas of two similar triangles equals the square of the ratio of corresponding sides.

Points to Remember (Exam Focus)

- ✓ Similarity criteria: AAA (or AA), SSS and SAS.
- ✓ AA is sufficient because the third angle follows from the angle sum property.
- ✓ Basic Proportionality Theorem: if $DE \parallel BC$ then $AD/DB = AE/EC$.
- ✓ Ratio of areas of similar triangles = (ratio of corresponding sides)².
- ✓ All congruent triangles are similar, but similar triangles need not be congruent.

⚠ Common Student Mistakes

- ⚠ Using SSA for similarity, which is not a valid criterion.
- ⚠ Taking the ratio of areas as the ratio of sides instead of its square.
- ⚠ Writing the correspondence of vertices in the wrong order.



Memory Hook

'Same shape, different size' — angles equal, sides proportional, areas in the squared ratio.



Advanced Insights

The squared area ratio generalises: for similar solids, volumes scale as the cube of the length ratio. This is why doubling every dimension of a model multiplies its weight by eight.

Formula Section



Basic Proportionality Theorem

Difficulty	● Medium ★★☆☆
Formula	If $DE \parallel BC$, then $AD/DB = AE/EC$
Meaning of Variables	D and E are points on AB and AC of triangle ABC.
When to Use	Whenever a line parallel to one side of a triangle cuts the other two sides.
Common Mistakes	Pairing the segments incorrectly — AD must go with AE, and DB with EC.
Shortcut / Memory Trick	The converse proves parallelism from the proportion.



Ratio of Areas of Similar Triangles

Difficulty	● Medium ★★☆☆
Formula	$\text{ar}(\triangle ABC) / \text{ar}(\triangle PQR) = (AB/PQ)^2 = (BC/QR)^2 = (CA/RP)^2$
Meaning of Variables	Corresponding sides of the two similar triangles.

When to Use	To find an unknown area or side ratio between two similar triangles.
Common Mistakes	Forgetting to square the side ratio.
Shortcut / Memory Trick	Sides in ratio 2:3 → areas in ratio 4:9.

6.2 Pythagoras Theorem and Its Converse

🕒 **Estimated time:** 14 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

Quick Summary

In a right triangle the square on the hypotenuse equals the sum of the squares on the other two sides, and the converse identifies right triangles from their side lengths.

Real-Life Example

A mason checks a corner is square using the 3-4-5 rule — the converse of the Pythagoras theorem applied with a piece of string.

Detailed Explanation

Definition

Pythagoras theorem: in a right-angled triangle, the square of the hypotenuse equals the sum of the squares of the other two sides. Its converse: if the square of one side equals the sum of the squares of the other two, the triangle is right-angled.

Key Idea

The modern proof uses similar triangles: the altitude from the right angle to the hypotenuse creates two triangles each similar to the original. The theorem follows in three lines.

Working Principle

The hypotenuse is always the longest side and lies opposite the right angle. Identifying it correctly before substituting is essential.

Applications

- Common Pythagorean triples: (3,4,5), (5,12,13), (8,15,17), (7,24,25).
- The distance formula in coordinate geometry is the Pythagoras theorem.
- Diagonal of a rectangle = $\sqrt{l^2 + b^2}$.

✓ Points to Remember (Exam Focus)

- ✓ In a right triangle: $\text{hypotenuse}^2 = \text{sum of squares of the other two sides}$.
- ✓ The hypotenuse is opposite the right angle and is the longest side.
- ✓ Converse: if $c^2 = a^2 + b^2$, the angle opposite c is a right angle.
- ✓ If $c^2 > a^2 + b^2$ the triangle is obtuse; if $c^2 < a^2 + b^2$ it is acute.

- ✓ The theorem is proved using similar triangles formed by the altitude to the hypotenuse.

⚠ Common Student Mistakes

- ⚠ Treating any side as the hypotenuse instead of the longest one.
- ⚠ Applying the theorem to a triangle that is not right-angled.
- ⚠ Forgetting to take the square root at the end.

🧠 Memory Hook

'Longest side squared equals the other two squared, added' — and 3-4-5 always works.

🚀 Advanced Insights

The theorem generalises to the cosine rule $c^2 = a^2 + b^2 - 2ab \cos C$, which reduces to Pythagoras when $C = 90^\circ$ and $\cos C = 0$. Pythagoras is simply the right-angled special case.

Formula Section

📐 Pythagoras Theorem

Difficulty	● Easy ★☆☆
Formula	$\text{hypotenuse}^2 = \text{base}^2 + \text{perpendicular}^2$
Meaning of Variables	hypotenuse = side opposite the right angle; the other two are the legs.
When to Use	For any right-angled triangle, and in distance and diagonal calculations.
Common Mistakes	Adding the squares of the hypotenuse and a leg.
Shortcut / Memory Trick	Recognise triples such as 3-4-5 and 5-12-13 to skip the arithmetic.

II Quick Recap — Topics Covered So Far

- Similar Triangles and Criteria for Similarity — Two triangles are similar if their corresponding angles are equal and corresponding sides are proportional, established by the AAA, SSS or SAS similarity criteria.
- Pythagoras Theorem and Its Converse — In a right triangle the square on the hypotenuse equals the sum of the squares on the other two sides, and the converse identifies right triangles from their side lengths.

Chapter 6 — End of Chapter Revision

All Memory Hooks

- Similar Triangles and Criteria for Similarity: 'Same shape, different size' — angles equal, sides proportional, areas in the squared ratio.
- Pythagoras Theorem and Its Converse: 'Longest side squared equals the other two squared, added' — and 3-4-5 always works.

All Common Mistakes

-  Using SSA for similarity, which is not a valid criterion.
-  Taking the ratio of areas as the ratio of sides instead of its square.
-  Writing the correspondence of vertices in the wrong order.
-  Treating any side as the hypotenuse instead of the longest one.
-  Applying the theorem to a triangle that is not right-angled.
-  Forgetting to take the square root at the end.

Formula Sheet

Formula	Meaning / Use
Basic Proportionality Theorem: If $DE \parallel BC$, then $AD/DB = AE/EC$	Whenever a line parallel to one side of a triangle cuts the other two sides.
Ratio of Areas of Similar Triangles: $\text{ar}(\triangle ABC) / \text{ar}(\triangle PQR) = (AB/PQ)^2 = (BC/QR)^2 = (CA/RP)^2$	To find an unknown area or side ratio between two similar triangles.
Pythagoras Theorem: $\text{hypotenuse}^2 = \text{base}^2 + \text{perpendicular}^2$	For any right-angled triangle, and in distance and diagonal calculations.

Quick Revision Section

- Similar Triangles and Criteria for Similarity: Two triangles are similar if their corresponding angles are equal and corresponding sides are proportional, established by the AAA, SSS or SAS similarity criteria.
- Pythagoras Theorem and Its Converse: In a right triangle the square on the hypotenuse equals the sum of the squares on the other two sides, and the converse identifies right triangles from their side lengths.

Chapter 7: Coordinate Geometry

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Plotting points in the Cartesian plane.
- Pythagoras theorem.
- Ratio and proportion.
- Properties of triangles and quadrilaterals.

7.1 Distance Formula and Its Applications

 **Estimated time:** 14 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

The distance between two points is $\sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$, which lets geometric properties be verified by calculation rather than by drawing.

Real-Life Example

A delivery app computes straight-line distance between two map coordinates with exactly this formula before choosing a route.

Detailed Explanation

Definition

The distance between points $A(x_1, y_1)$ and $B(x_2, y_2)$ is $AB = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$. It is the Pythagoras theorem applied to the horizontal and vertical differences.

Key Idea

Because the differences are squared, the order of subtraction does not matter. Distance is therefore always positive, whichever point is taken first.

Working Principle

To classify a figure, compute all sides and both diagonals. Equal sides with equal diagonals give a square; equal sides with unequal diagonals give a rhombus.

Applications

- Collinearity: three points are collinear if the sum of two distances equals the third.
- Circumcentre: the point equidistant from all three vertices of a triangle.
- Type of triangle: equilateral, isosceles, scalene or right-angled, from the side lengths.

✓ **Points to Remember (Exam Focus)**

- ✓ Distance = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.
- ✓ Distance from the origin to $(x, y) = \sqrt{x^2 + y^2}$.
- ✓ Equal sides and equal diagonals $\rightarrow$ square; equal sides, unequal diagonals $\rightarrow$ rhombus.
- ✓ Equal diagonals with unequal adjacent sides $\rightarrow$ rectangle.
- ✓ Three points are collinear if $AB + BC = AC$ for some ordering.

⚠ Common Student Mistakes

- ⚠ Subtracting the coordinates in mixed order, e.g. $(x_2 - x_1)$ with $(y_1 - y_2)$.
- ⚠ Forgetting to square the differences before adding.
- ⚠ Concluding a figure is a square from equal sides alone, without checking the diagonals.



Memory Hook

'Difference of x, difference of y, square, add, root' — Pythagoras in coordinates.



Advanced Insights

The formula extends unchanged to three dimensions by adding $(z_2 - z_1)^2$, and to n dimensions by adding one term per coordinate. It is the Euclidean metric that defines distance in any dimension.

Formula Section



Distance Formula

Difficulty	● Easy ★☆☆
Formula	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Meaning of Variables	(x_1, y_1) and (x_2, y_2) = the two points.
When to Use	For lengths of sides and diagonals, and for classifying figures.
Common Mistakes	Mixing the order of subtraction between the x and y terms.
Shortcut / Memory Trick	The order does not matter because the differences are squared.

7.2 Section Formula and Area of a Triangle

🕒 Estimated time: 15 mins Difficulty: ● Medium ★★☆☆ Exam Importance: ★★★★★



Quick Summary

The section formula locates the point dividing a segment in a given ratio, and the area of a triangle can be found from the coordinates of its vertices alone.



Real-Life Example

Placing a milestone exactly two-fifths of the way along a road between two towns is a section formula calculation.



Detailed Explanation

Definition

The point P dividing the join of $A(x_1, y_1)$ and $B(x_2, y_2)$ internally in the ratio $m_1 : m_2$ has coordinates $[(m_1x_2 + m_2x_1)/(m_1 + m_2), (m_1y_2 + m_2y_1)/(m_1 + m_2)]$.

Key Idea

The cross-pairing in the section formula is what students most often get wrong: m_1 multiplies x_2 , not x_1 . Checking the formula against the mid-point case ($m_1 = m_2$) catches the error instantly.

Working Principle

The area formula gives zero exactly when the three points are collinear, which makes it the standard test for collinearity.

Applications

- Mid-point formula: $[(x_1 + x_2)/2, (y_1 + y_2)/2]$.
- Centroid of a triangle: $[(x_1 + x_2 + x_3)/3, (y_1 + y_2 + y_3)/3]$.
- Area zero $\rightarrow$ the three points are collinear.

✓ Points to Remember (Exam Focus)

- ✓ Section formula: $[(m_1x_2 + m_2x_1)/(m_1 + m_2), (m_1y_2 + m_2y_1)/(m_1 + m_2)]$.
- ✓ Mid-point is the special case $m_1 : m_2 = 1 : 1$.
- ✓ Centroid divides each median in the ratio 2 : 1 from the vertex.
- ✓ Area = $\frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$.
- ✓ Area = 0 is the test for collinearity.

⚠ Common Student Mistakes

- ⚠ Pairing m_1 with x_1 instead of x_2 in the section formula.
- ⚠ Omitting the modulus and reporting a negative area.
- ⚠ Reversing the ratio when the question specifies the order of the points.

 Memory Hook

'Cross the ratio over: m_1 goes with x_2 ' — and check it reduces to the mid-point when $m_1 = m_2$.

 Advanced Insights

The area formula is the determinant of a 3×3 matrix of the coordinates, halved. Determinants therefore measure area in two dimensions and volume in three — one idea underlying both.

II Quick Recap — Topics Covered So Far

- Distance Formula and Its Applications — The distance between two points is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, which lets geometric properties be verified by calculation rather than by drawing.
- Section Formula and Area of a Triangle — The section formula locates the point dividing a segment in a given ratio, and the area of a triangle can be found from the coordinates of its vertices alone.

Chapter 7 — End of Chapter Revision

All Memory Hooks

- Distance Formula and Its Applications: 'Difference of x, difference of y, square, add, root' — Pythagoras in coordinates.
- Section Formula and Area of a Triangle: 'Cross the ratio over: m_1 goes with x_2 ' — and check it reduces to the mid-point when $m_1 = m_2$.

All Common Mistakes

-  Subtracting the coordinates in mixed order, e.g. $(x_2 - x_1)$ with $(y_1 - y_2)$.
-  Forgetting to square the differences before adding.
-  Concluding a figure is a square from equal sides alone, without checking the diagonals.
-  Pairing m_1 with x_1 instead of x_2 in the section formula.
-  Omitting the modulus and reporting a negative area.
-  Reversing the ratio when the question specifies the order of the points.

Formula Sheet

Formula	Meaning / Use
Distance Formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	For lengths of sides and diagonals, and for classifying figures.

Quick Revision Section

- Distance Formula and Its Applications: The distance between two points is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, which lets geometric properties be verified by calculation rather than by drawing.
- Section Formula and Area of a Triangle: The section formula locates the point dividing a segment in a given ratio, and the area of a triangle can be found from the coordinates of its vertices alone.

Chapter 8: Introduction to Trigonometry

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Pythagoras theorem and right-angled triangles.
- Similarity of triangles.
- Ability to simplify algebraic fractions.
- Familiarity with surds such as $\sqrt{2}$ and $\sqrt{3}$.

8.1 Trigonometric Ratios and Standard Angles

 **Estimated time:** 16 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

The six trigonometric ratios relate the angles of a right triangle to its sides, and their values at 0° , 30° , 45° , 60° and 90° must be known by heart.

Real-Life Example

A ramp's steepness is quoted as a gradient, which is exactly the tangent of the angle it makes with the ground.

Detailed Explanation

Definition

For an acute angle θ in a right triangle: $\sin \theta = \text{opposite/hypotenuse}$, $\cos \theta = \text{adjacent/hypotenuse}$, $\tan \theta = \text{opposite/adjacent}$, and cosec, sec and cot are their reciprocals.

Key Idea

The ratios depend only on the angle, not on the size of the triangle, because all right triangles with the same acute angle are similar. This is why a single table of values serves every triangle.

Working Principle

The standard-angle table is generated from two triangles: the 45-45-90 triangle from a square's diagonal, and the 30-60-90 triangle from an equilateral triangle cut in half.

Applications

- $\sin \theta$ increases from 0 to 1 as θ goes from 0° to 90° ; $\cos \theta$ decreases from 1 to 0.
- $\tan 90^\circ$ and $\sec 90^\circ$ are undefined; $\cot 0^\circ$ and $\text{cosec } 0^\circ$ are undefined.
- $\sin \theta$ and $\cos \theta$ can never exceed 1 in magnitude.

Points to Remember (Exam Focus)

- ✓ $\sin \theta = P/H$, $\cos \theta = B/H$, $\tan \theta = P/B$; cosec, sec, cot are the reciprocals.
- ✓ $\sin 0^\circ = 0$, $\sin 30^\circ = \frac{1}{2}$, $\sin 45^\circ = \frac{1}{\sqrt{2}}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\sin 90^\circ = 1$.
- ✓ cos values run in exactly the reverse order of sin values.
- ✓ $\tan \theta = \sin \theta / \cos \theta$; $\cot \theta = \cos \theta / \sin \theta$.
- ✓ $\sin \theta$ and $\cos \theta$ always lie between -1 and 1 .

⚠ Common Student Mistakes

- ⚠ Interchanging sin and cos when the angle is measured from the other vertex.
- ⚠ Writing $\sin 90^\circ = 0$ instead of 1 .
- ⚠ Treating $\tan 90^\circ$ as infinity in a proof; it is undefined.



Memory Hook

'SOH-CAH-TOA' — and for the table, sin runs $0, \frac{1}{2}, \frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{2}, 1$ while cos runs backwards.



Advanced Insights

Extending these ratios to angles beyond 90° needs the unit circle, where sine and cosine become the coordinates of a point. That extension turns trigonometry from triangle measurement into the mathematics of waves.

Formula Section



Trigonometric Ratios

Difficulty	● Easy ★☆☆
Formula	$\sin \theta = P/H$, $\cos \theta = B/H$, $\tan \theta = P/B$, $\operatorname{cosec} \theta = H/P$, $\sec \theta = H/B$, $\cot \theta = B/P$
Meaning of Variables	P = perpendicular (opposite side), B = base (adjacent side), H = hypotenuse.
When to Use	For any right-angled triangle where an angle and one side are known.
Common Mistakes	Identifying the wrong side as the perpendicular when the angle changes.
Shortcut / Memory Trick	The side opposite the angle is always the perpendicular for that angle.

8.2 Trigonometric Identities

🕒 Estimated time: 15 mins Difficulty: ● Medium ★★☆☆ Exam Importance: ★★★★★

Quick Summary

The three Pythagorean identities relate the squares of the trigonometric ratios and are the tools for proving every other trigonometric result.

Real-Life Example

Signal processing constantly uses $\sin^2\theta + \cos^2\theta = 1$ to convert between amplitude and power without knowing the angle.

Detailed Explanation

Definition

An identity is an equation true for every value of the variable for which both sides are defined. The fundamental identity is $\sin^2\theta + \cos^2\theta = 1$.

Key Idea

All three identities are the Pythagoras theorem divided by different sides. Dividing $\sin^2\theta + \cos^2\theta = 1$ by $\cos^2\theta$ gives the sec identity, and by $\sin^2\theta$ gives the cosec identity.

Working Principle

The standard proof technique is to convert everything to sines and cosines, simplify, and compare. Working on one side only, and never across the equality sign, is what examiners expect.

Applications

- Complementary angles: $\sin(90^\circ - \theta) = \cos \theta$ and $\tan(90^\circ - \theta) = \cot \theta$.
- Rationalising expressions such as $(1 + \sin \theta)/\cos \theta$ using the identities.
- Eliminating θ between two given equations.

✓ Points to Remember (Exam Focus)

- ✓ $\sin^2\theta + \cos^2\theta = 1$.
- ✓ $1 + \tan^2\theta = \sec^2\theta$.
- ✓ $1 + \cot^2\theta = \operatorname{cosec}^2\theta$.
- ✓ $\sin(90^\circ - \theta) = \cos \theta$, $\cos(90^\circ - \theta) = \sin \theta$, $\tan(90^\circ - \theta) = \cot \theta$.
- ✓ Work on one side of an identity only; never operate across the equals sign.

Common Student Mistakes

-  Writing $\sin^2\theta$ to mean $\sin(\theta^2)$; it means $(\sin \theta)^2$.
-  Using $\sec^2\theta = 1 - \tan^2\theta$ instead of $1 + \tan^2\theta$.
-  Cross-multiplying across the equality while proving an identity.

 Memory Hook

'Divide the first identity by $\cos^2$ for sec, by $\sin^2$ for cosec' — three identities, one source.

 Advanced Insights

$\sin^2\theta + \cos^2\theta = 1$ is the equation of the unit circle in disguise. Every trigonometric identity is ultimately a statement about points on that circle.

Formula Section **Fundamental Trigonometric Identities**

Difficulty	● Medium ★★☆☆
Formula	$\sin^2\theta + \cos^2\theta = 1$; $1 + \tan^2\theta = \sec^2\theta$; $1 + \cot^2\theta = \operatorname{cosec}^2\theta$
Meaning of Variables	θ = any angle for which the ratios are defined.
When to Use	To prove identities, simplify expressions and eliminate variables.
Common Mistakes	Getting the sign wrong: it is $1 + \tan^2\theta$, not $1 - \tan^2\theta$.
Shortcut / Memory Trick	Convert everything to sin and cos when a proof stalls.

II Quick Recap — Topics Covered So Far

- **Trigonometric Ratios and Standard Angles** — The six trigonometric ratios relate the angles of a right triangle to its sides, and their values at 0° , 30° , 45° , 60° and 90° must be known by heart.
- **Trigonometric Identities** — The three Pythagorean identities relate the squares of the trigonometric ratios and are the tools for proving every other trigonometric result.

Chapter 8 — End of Chapter Revision

All Memory Hooks

- Trigonometric Ratios and Standard Angles: 'SOH-CAH-TOA' — and for the table, sin runs 0, $\frac{1}{2}$, $\frac{1}{\sqrt{2}}$, $\frac{\sqrt{3}}{2}$, 1 while cos runs backwards.
- Trigonometric Identities: 'Divide the first identity by $\cos^2$ for sec, by $\sin^2$ for cosec' — three identities, one source.

All Common Mistakes

-  Interchanging sin and cos when the angle is measured from the other vertex.
-  Writing $\sin 90^\circ = 0$ instead of 1.
-  Treating $\tan 90^\circ$ as infinity in a proof; it is undefined.
-  Writing $\sin^2\theta$ to mean $\sin(\theta^2)$; it means $(\sin \theta)^2$.
-  Using $\sec^2\theta = 1 - \tan^2\theta$ instead of $1 + \tan^2\theta$.
-  Cross-multiplying across the equality while proving an identity.

Formula Sheet

Formula	Meaning / Use
Trigonometric Ratios: $\sin \theta = P/H$, $\cos \theta = B/H$, $\tan \theta = P/B$, $\operatorname{cosec} \theta = H/P$, $\sec \theta = H/B$, $\cot \theta = B/P$	For any right-angled triangle where an angle and one side are known.
Fundamental Trigonometric Identities: $\sin^2\theta + \cos^2\theta = 1$; $1 + \tan^2\theta = \sec^2\theta$; $1 + \cot^2\theta = \operatorname{cosec}^2\theta$	To prove identities, simplify expressions and eliminate variables.

Quick Revision Section

- Trigonometric Ratios and Standard Angles: The six trigonometric ratios relate the angles of a right triangle to its sides, and their values at 0° , 30° , 45° , 60° and 90° must be known by heart.
- Trigonometric Identities: The three Pythagorean identities relate the squares of the trigonometric ratios and are the tools for proving every other trigonometric result.

Chapter 9: Some Applications of Trigonometry

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Trigonometric ratios of standard angles.
- Ability to draw and label a right-angled triangle.
- Skill in solving linear equations with surds.
- Understanding of the horizontal and the vertical.

9.1 Heights and Distances

 **Estimated time:** 16 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

The angle of elevation and the angle of depression allow inaccessible heights and distances to be computed from a single measured length.

Real-Life Example

The height of a tower is found by measuring the angle from a known distance away — no one has to climb anything.

Detailed Explanation

Definition

The angle of elevation is the angle between the horizontal and the line of sight to an object above the observer. The angle of depression is the angle between the horizontal and the line of sight to an object below.

Key Idea

The angle of depression from a tower to a point equals the angle of elevation from that point to the tower, because they are alternate angles between parallel horizontals. Redrawing the figure using this equality is the standard first step.

Working Principle

Draw a clear figure, mark the right angle and all given data, choose the ratio that connects the known side with the required one, and solve. The choice of ratio is the whole of the technique.

Applications

- $\tan$ is used when the two sides involved are the perpendicular and the base.
- $\sin$ or $\cos$ is used whenever the hypotenuse is involved.
- The observer's height must be added when the angle is measured from eye level.

✓ Points to Remember (Exam Focus)

- ✓ Angle of elevation: looking up. Angle of depression: looking down.
- ✓ Angle of depression = angle of elevation from the other end (alternate angles).
- ✓ The line of sight is the hypotenuse; the horizontal is the base.
- ✓ Choose tan when the hypotenuse is neither known nor required.
- ✓ Add the observer's height when the angle is measured from the eye.

⚠ Common Student Mistakes

- ⚠ Marking the angle of depression inside the triangle instead of with the horizontal.
- ⚠ Forgetting to add the height of the observer to the final answer.
- ⚠ Using sin where tan is required, which brings in an unknown hypotenuse.



Memory Hook

'Depression is measured from the horizontal, never from the vertical' — draw the horizontal line first.



Advanced Insights

Beyond right triangles, the sine and cosine rules handle oblique triangles and are the basis of triangulation. Land surveying and satellite positioning both extend exactly this method.

Formula Section



Heights and Distances

Difficulty	● Medium ★★☆☆
Formula	$\tan \theta = \text{height} / \text{horizontal distance}$
Meaning of Variables	θ = angle of elevation or depression, height = vertical, distance = horizontal.
When to Use	Whenever a vertical height and a horizontal distance are involved and the hypotenuse is not needed.
Common Mistakes	Interchanging the height and the distance in the ratio.
Shortcut / Memory Trick	$\tan 45^\circ = 1$, so at 45° the height equals the horizontal distance exactly.

II Quick Recap — Topics Covered So Far

- Heights and Distances — The angle of elevation and the angle of depression allow inaccessible heights and distances to be computed from a single measured length.

Chapter 9 — End of Chapter Revision

All Memory Hooks

- Heights and Distances: 'Depression is measured from the horizontal, never from the vertical' — draw the horizontal line first.

All Common Mistakes

-  Marking the angle of depression inside the triangle instead of with the horizontal.
-  Forgetting to add the height of the observer to the final answer.
-  Using sin where tan is required, which brings in an unknown hypotenuse.

Formula Sheet

Formula	Meaning / Use
Heights and Distances: $\tan \theta = \text{height} / \text{horizontal distance}$	Whenever a vertical height and a horizontal distance are involved and the hypotenuse is not needed.

Quick Revision Section

- Heights and Distances: The angle of elevation and the angle of depression allow inaccessible heights and distances to be computed from a single measured length.

Chapter 10: Circles

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Chord and angle properties of circles from Class 9.
- Congruence of triangles.
- Pythagoras theorem.
- Idea of a line touching a curve.

10.1 Tangent to a Circle and Its Properties

 **Estimated time:** 15 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

A tangent touches a circle at exactly one point and is perpendicular to the radius at that point, and the two tangents from an external point are equal in length.

Real-Life Example

A bicycle chain leaves each sprocket along a tangent, and the two straight stretches from a common point are always of equal length.

Detailed Explanation

Definition

A tangent to a circle is a line that intersects the circle at exactly one point, called the point of contact. A secant intersects the circle at two points.

Key Idea

The tangent–radius perpendicularity is the key to every problem in this chapter. Drawing the radius to the point of contact immediately creates a right triangle to work with.

Working Principle

From an external point P, the two tangent lengths PA and PB are equal because triangles OAP and OBP are congruent by RHS, sharing OP and having equal radii.

Applications

- No tangent can be drawn from a point inside the circle; exactly one from a point on it; exactly two from an external point.
- The length of a tangent from an external point is $\sqrt{d^2 - r^2}$, where d is the distance from the centre.
- OP bisects both the angle between the tangents and the angle at the centre.

✓ Points to Remember (Exam Focus)

- ✓ A tangent is perpendicular to the radius at the point of contact.
- ✓ Tangents drawn from an external point are equal in length.
- ✓ A circle has infinitely many tangents, one at each point.
- ✓ Tangent length from an external point = $\sqrt{d^2 - r^2}$.
- ✓ The line joining the external point to the centre bisects the angle between the tangents.

⚠ Common Student Mistakes

- ⚠ Assuming the tangent is perpendicular to the chord rather than to the radius.
- ⚠ Trying to draw a tangent from a point inside the circle.
- ⚠ Using the radius in place of the distance from the centre in the tangent-length formula.



Memory Hook

'Radius meets tangent at a right angle' — draw that radius and a right triangle appears.



Advanced Insights

A tangent is the limiting position of a secant as the two intersection points merge. This limiting idea is precisely what a derivative captures in calculus.

Formula Section



Length of a Tangent

Difficulty	● Medium ★★☆☆
Formula	Tangent length = $\sqrt{d^2 - r^2}$
Meaning of Variables	d = distance of the external point from the centre, r = radius of the circle.
When to Use	To find the tangent length from a point outside the circle.
Common Mistakes	Adding instead of subtracting under the root.
Shortcut / Memory Trick	It is just Pythagoras on the right triangle formed by radius, tangent and the line to the centre.

II Quick Recap — Topics Covered So Far

- **Tangent to a Circle and Its Properties** — A tangent touches a circle at exactly one point and is perpendicular to the radius at that point, and the two tangents from an external point are equal in length.

Chapter 10 — End of Chapter Revision

All Memory Hooks

- Tangent to a Circle and Its Properties: 'Radius meets tangent at a right angle' — draw that radius and a right triangle appears.

All Common Mistakes

-  Assuming the tangent is perpendicular to the chord rather than to the radius.
-  Trying to draw a tangent from a point inside the circle.
-  Using the radius in place of the distance from the centre in the tangent-length formula.

Formula Sheet

Formula	Meaning / Use
Length of a Tangent: $\text{Tangent length} = \sqrt{d^2 - r^2}$	To find the tangent length from a point outside the circle.

Quick Revision Section

- Tangent to a Circle and Its Properties: A tangent touches a circle at exactly one point and is perpendicular to the radius at that point, and the two tangents from an external point are equal in length.

Chapter 11: Constructions

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Basic ruler-and-compass constructions from Class 9.
- Basic Proportionality Theorem.
- Similarity of triangles and the ratio of sides.
- Tangent properties of a circle.

11.1 Division of a Segment and Similar Triangles

 **Estimated time:** 14 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**    

Quick Summary

A line segment can be divided in any given ratio using a ray of equally spaced points, and the same idea constructs a triangle similar to a given one in a required scale factor.

Real-Life Example

An architect scaling a plan to two-thirds size uses exactly this construction rather than recomputing every measurement.

Detailed Explanation

Definition

To divide a segment AB in the ratio $m : n$, draw an acute-angled ray from A , mark $m + n$ equal arcs along it, join the last point to B , and draw a parallel through the m th point.

Key Idea

The construction works by the Basic Proportionality Theorem — the parallel line divides AB in the same ratio as the ray was divided. Every step has a justification.

Working Principle

For a similar triangle with scale factor m/n , mark the larger of m and n arcs along the ray. Getting this choice right is what decides whether the new triangle is larger or smaller.

Applications

- Scale factor greater than 1 gives a larger triangle; less than 1 gives a smaller one.
- The arcs on the ray must be exactly equal, so the compass width must not change.
- The justification is expected in the answer, not only the drawing.

✓ **Points to Remember (Exam Focus)**

- ✓ Mark $m + n$ equal arcs when dividing a segment in the ratio $m : n$.
- ✓ The construction is justified by the Basic Proportionality Theorem.
- ✓ For a scale factor m/n , use the greater of m and n as the number of arcs.
- ✓ Keep the compass width unchanged while marking equal arcs.
- ✓ Write the steps of construction and the justification; both carry marks.

⚠ Common Student Mistakes

- ⚠ Marking m arcs instead of $m + n$ when dividing in the ratio $m : n$.
- ⚠ Changing the compass width partway through marking the arcs.
- ⚠ Omitting the justification, which is separately marked.

🧠 Memory Hook

' $m + n$ equal arcs, then one parallel line' — the ratio transfers automatically.

🚀 Advanced Insights

This is a constructive proof of the Basic Proportionality Theorem. The same principle underlies the slide rule and every mechanical scaling device built before calculators.

11.2 Construction of Tangents to a Circle

🕒 **Estimated time:** 13 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

🔑 Quick Summary

Tangents from an external point are constructed by drawing a circle on the line joining the point to the centre as diameter, and joining the intersections.

🌐 Real-Life Example

Marking where a belt leaves a pulley when it runs to a fixed post is exactly this construction on a workshop floor.

📖 Detailed Explanation

Definition

To construct tangents from an external point P to a circle with centre O , find the mid-point M of OP , draw a circle with centre M and radius MO , and join P to the two points where the circles intersect.

Key Idea

The construction works because an angle in a semicircle is a right angle. The circle on OP as diameter forces $OA \perp PA$, which is exactly the tangent condition.

Working Principle

When the centre is not marked, it is first located by drawing two chords and their perpendicular bisectors, which meet at the centre.

Applications

- Two tangents can be drawn from any external point.
- Only one tangent exists at a point on the circle itself.
- The tangent lengths from P are equal, which serves as a check on the drawing.

✓ Points to Remember (Exam Focus)

- ✓ Find the mid-point of OP and draw a circle with OP as diameter.
- ✓ The intersections of the two circles are the points of contact.
- ✓ Justification: angle in a semicircle is 90° , so the radius is perpendicular to the tangent.
- ✓ If the centre is not given, locate it using the perpendicular bisectors of two chords.
- ✓ Check the drawing by verifying that the two tangent lengths are equal.

⚠ Common Student Mistakes

- ⚠ Taking the mid-point of the radius instead of OP .
- ⚠ Drawing tangents by eye rather than by construction.
- ⚠ Omitting the justification based on the angle in a semicircle.

🧠 Memory Hook

'Circle on OP as diameter' — the semicircle supplies the right angle for free.

🚀 Advanced Insights

The construction is the geometric solution of a quadratic: the two intersection points correspond to the two roots. Where the external point lies on the circle, the roots coincide and only one tangent exists.

II Quick Recap — Topics Covered So Far

- Division of a Segment and Similar Triangles — A line segment can be divided in any given ratio using a ray of equally spaced points, and the same idea constructs a triangle similar to a given one in a required scale factor.

- Construction of Tangents to a Circle — Tangents from an external point are constructed by drawing a circle on the line joining the point to the centre as diameter, and joining the intersections.

Chapter 11 — End of Chapter Revision

All Memory Hooks

- Division of a Segment and Similar Triangles: 'm + n equal arcs, then one parallel line' — the ratio transfers automatically.
- Construction of Tangents to a Circle: 'Circle on OP as diameter' — the semicircle supplies the right angle for free.

All Common Mistakes

-  Marking m arcs instead of m + n when dividing in the ratio m : n.
-  Changing the compass width partway through marking the arcs.
-  Omitting the justification, which is separately marked.
-  Taking the mid-point of the radius instead of OP.
-  Drawing tangents by eye rather than by construction.
-  Omitting the justification based on the angle in a semicircle.

Quick Revision Section

- Division of a Segment and Similar Triangles: A line segment can be divided in any given ratio using a ray of equally spaced points, and the same idea constructs a triangle similar to a given one in a required scale factor.
- Construction of Tangents to a Circle: Tangents from an external point are constructed by drawing a circle on the line joining the point to the centre as diameter, and joining the intersections.

Chapter 12: Areas Related to Circles

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Formulae for circumference and area of a circle.
- Value of π as $22/7$ or 3.14 .
- Area of a triangle, including the equilateral case.
- Understanding of an angle measured in degrees.

12.1 Sector and Segment of a Circle

 **Estimated time:** 16 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

A sector is the region between two radii and an arc, and a segment is the region between a chord and an arc; both are found from the central angle as a fraction of 360° .

Real-Life Example

A slice of pizza is a sector, and the crust-side piece left after a straight cut across it is a segment.

Detailed Explanation

Definition

A sector is the region enclosed by two radii and the arc between them. A segment is the region enclosed by a chord and the corresponding arc. The minor sector or segment is the smaller of the two parts.

Key Idea

Every formula in the chapter is the whole-circle formula multiplied by $\theta/360$. Once that is seen, only the area and circumference of a circle need to be remembered.

Working Principle

Area of a segment = area of the sector – area of the triangle formed by the two radii and the chord. Choosing the right triangle formula for that step is where most marks are lost.

Applications

- Area swept by a clock hand or a windscreen wiper is a sector area.
- Grazing area of an animal tied at the corner of a field is a sector.
- Designs combining circles and squares are handled by adding and subtracting areas.

Points to Remember (Exam Focus)

- ✓ Length of arc = $(\theta/360) \times 2\pi r$.
- ✓ Area of sector = $(\theta/360) \times \pi r^2$.
- ✓ Area of segment = area of sector – area of the triangle.
- ✓ For a 60° sector the triangle is equilateral, with area $(\sqrt{3}/4)r^2$.
- ✓ Area of sector can also be written as $\frac{1}{2} \times \text{arc length} \times r$.

⚠ Common Student Mistakes

- ⚠ Using $\theta/180$ instead of $\theta/360$ in the sector formulas.
- ⚠ Forgetting to subtract the triangle when the segment is required.
- ⚠ Using the diameter in place of the radius.



Memory Hook

'Sector = slice of pizza; Segment = sector minus triangle.'



Advanced Insights

In radian measure the formulas simplify to arc = $r\theta$ and sector area = $\frac{1}{2}r^2\theta$, with no 360 anywhere. That simplification is the whole reason radians are the standard unit in higher mathematics.

Formula Section



Arc Length and Sector Area

Difficulty	● Medium ★★☆☆
Formula	Arc = $(\theta/360) \cdot 2\pi r$; Sector area = $(\theta/360) \cdot \pi r^2 = \frac{1}{2} \times \text{arc} \times r$
Meaning of Variables	θ = central angle in degrees, r = radius.
When to Use	For any sector, wiper, grazing or clock-hand problem.
Common Mistakes	Dividing by 180 instead of 360.
Shortcut / Memory Trick	$\theta = 90^\circ$ gives a quarter circle; $\theta = 180^\circ$ gives a semicircle.



Area of a Segment

Difficulty	● Medium ★★☆☆
Formula	Segment area = $(\theta/360) \cdot \pi r^2 - \frac{1}{2}r^2 \sin \theta$
Meaning of Variables	θ = central angle, r = radius.

When to Use	For the region between a chord and its arc.
Common Mistakes	Forgetting to subtract the triangular part.
Shortcut / Memory Trick	For $\theta = 60^\circ$ the triangle is equilateral with area $(\sqrt{3}/4)r^2$.

II Quick Recap — Topics Covered So Far

- **Sector and Segment of a Circle** — A sector is the region between two radii and an arc, and a segment is the region between a chord and an arc; both are found from the central angle as a fraction of 360° .

Chapter 12 — End of Chapter Revision

All Memory Hooks

- Sector and Segment of a Circle: 'Sector = slice of pizza; Segment = sector minus triangle.'

All Common Mistakes

-  Using $\theta/180$ instead of $\theta/360$ in the sector formulas.
-  Forgetting to subtract the triangle when the segment is required.
-  Using the diameter in place of the radius.

Formula Sheet

Formula	Meaning / Use
Arc Length and Sector Area: $\text{Arc} = (\theta/360) \cdot 2\pi r$; Sector area = $(\theta/360) \cdot \pi r^2 = \frac{1}{2} \times \text{arc} \times r$	For any sector, wiper, grazing or clock-hand problem.
Area of a Segment: Segment area = $(\theta/360) \cdot \pi r^2 - \frac{1}{2}r^2 \sin \theta$	For the region between a chord and its arc.

Quick Revision Section

- Sector and Segment of a Circle: A sector is the region between two radii and an arc, and a segment is the region between a chord and an arc; both are found from the central angle as a fraction of 360° .

Chapter 13: Surface Areas and Volumes

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Surface area and volume formulae for cuboid, cylinder, cone and sphere.
- Value of π and skill with fractions.
- Pythagoras theorem for finding slant height.
- Ratio and proportion.

13.1 Combination of Solids

 **Estimated time:** 16 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

For a solid made by joining two shapes, volumes always add, but only the exposed surfaces are counted in the surface area.

Real-Life Example

A capsule is a cylinder with a hemisphere at each end, and a circus tent is a cylinder topped by a cone — both standard combination problems.

Detailed Explanation

Definition

A combination of solids is formed by joining two or more standard solids, or by removing one from another. Its volume is the sum or difference of the component volumes.

Key Idea

Surfaces that are joined together disappear from view and must not be counted. Volumes, by contrast, always add without exception — this asymmetry is the heart of the chapter.

Working Principle

Identify the component solids, decide which faces remain exposed, add only those areas, and add all the volumes.

Applications

- Cylinder + hemisphere: total surface area = $2\pi rh + 2\pi r^2 + \pi r^2$ (base), if the base is exposed.
- Cone on a cylinder (a tent): canvas needed = curved surface of cone + curved surface of cylinder.
- A cylinder with a conical cavity: subtract the cone's volume but add its curved surface.

Points to Remember (Exam Focus)

- ✓ Volumes always add; only exposed surface areas add.
- ✓ The circular faces where two solids join are not part of the surface area.
- ✓ Draw the solid and mark which faces are visible before writing any formula.
- ✓ For a cone, the curved surface uses slant height $l = \sqrt{r^2 + h^2}$.
- ✓ Check that r is the same for both parts where they join.

⚠ Common Student Mistakes

- ⚠ Including the joined circular faces in the total surface area.
- ⚠ Using the vertical height instead of the slant height for a cone's curved surface.
- ⚠ Subtracting volumes when the solids are joined rather than hollowed out.

🧠 Memory Hook

'Add all volumes, but only the surfaces you can actually see.'

🚀 Advanced Insights

For a hollowed solid the surface area may increase even though the volume decreases, because a new internal surface is created. Volume and surface area are genuinely independent quantities.

13.2 Conversion of Solids and Frustum of a Cone

🕒 **Estimated time:** 15 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

🔑 Quick Summary

When a solid is melted and recast the volume stays constant, and a frustum is the part of a cone left after cutting off the top parallel to the base.

🌐 Real-Life Example

A bucket, a lampshade and a drinking glass are all frustums, and melting a metal sphere into wire is a volume-conservation problem.

📖 Detailed Explanation

Definition

A frustum of a cone is the solid remaining when a cone is cut by a plane parallel to its base and the upper cone is removed. Its slant height is $l = \sqrt{h^2 + (R - r)^2}$.

Key Idea

In every melting, recasting or reshaping problem the volume is conserved while the surface area changes freely. Setting the two volumes equal is always the first line of the solution.

Working Principle

The frustum formulas come from subtracting the small cone from the large one; the algebra is done once and the result quoted.

Applications

- Number of small articles from a large one = volume of large $\div$ volume of one small.
- A bucket's capacity is a frustum volume; the metal sheet needed is its curved surface plus the base.
- Water flowing through a pipe: volume = cross-sectional area $\times$ speed $\times$ time.

✓ Points to Remember (Exam Focus)

- ✓ In melting and recasting, volume is conserved but surface area is not.
- ✓ Frustum volume = $(1/3)\pi h(R^2 + r^2 + Rr)$.
- ✓ Frustum curved surface = $\pi l(R + r)$, with $l = \sqrt{h^2 + (R - r)^2}$.
- ✓ Frustum total surface = $\pi l(R + r) + \pi R^2 + \pi r^2$.
- ✓ For a flowing pipe, volume delivered = area of cross-section $\times$ flow speed $\times$ time.

⚠ Common Student Mistakes

- ⚠ Using $l = \sqrt{h^2 + R^2 + r^2}$ instead of $\sqrt{h^2 + (R - r)^2}$.
- ⚠ Equating surface areas instead of volumes in a recasting problem.
- ⚠ Forgetting that a bucket usually has no lid, so the top circle is not included.

🧠 Memory Hook

'Melt and recast: volume never changes' — and for a frustum, use $(R - r)$ in the slant height.

🚀 Advanced Insights

Letting $r \rightarrow 0$ turns the frustum formula into the cone formula, and $r \rightarrow R$ turns it into the cylinder formula. A single expression contains both as limiting cases.

Formula Section**📐 Frustum of a Cone**

Difficulty	● Hard ★★★
Formula	$V = (1/3)\pi h(R^2 + r^2 + Rr)$; $CSA = \pi l(R + r)$; $l = \sqrt{h^2 + (R - r)^2}$

Meaning of Variables	R = larger radius, r = smaller radius, h = vertical height, l = slant height.
When to Use	For buckets, glasses, lampshades and any truncated cone.
Common Mistakes	Adding R and r inside the slant-height root instead of subtracting.
Shortcut / Memory Trick	Setting $r = 0$ recovers the cone formulas as a check.

II Quick Recap — Topics Covered So Far

- **Combination of Solids** — For a solid made by joining two shapes, volumes always add, but only the exposed surfaces are counted in the surface area.
- **Conversion of Solids and Frustum of a Cone** — When a solid is melted and recast the volume stays constant, and a frustum is the part of a cone left after cutting off the top parallel to the base.

Chapter 13 — End of Chapter Revision

All Memory Hooks

- Combination of Solids: 'Add all volumes, but only the surfaces you can actually see.'
- Conversion of Solids and Frustum of a Cone: 'Melt and recast: volume never changes' — and for a frustum, use $(R - r)$ in the slant height.

All Common Mistakes

-  Including the joined circular faces in the total surface area.
-  Using the vertical height instead of the slant height for a cone's curved surface.
-  Subtracting volumes when the solids are joined rather than hollowed out.
-  Using $l = \sqrt{h^2 + R^2 + r^2}$ instead of $\sqrt{h^2 + (R - r)^2}$.
-  Equating surface areas instead of volumes in a recasting problem.
-  Forgetting that a bucket usually has no lid, so the top circle is not included.

Formula Sheet

Formula	Meaning / Use
Frustum of a Cone: $V = (1/3)\pi h(R^2 + r^2 + Rr)$; $CSA = \pi l(R + r)$; $l = \sqrt{h^2 + (R - r)^2}$	For buckets, glasses, lampshades and any truncated cone.

Quick Revision Section

- Combination of Solids: For a solid made by joining two shapes, volumes always add, but only the exposed surfaces are counted in the surface area.
- Conversion of Solids and Frustum of a Cone: When a solid is melted and recast the volume stays constant, and a frustum is the part of a cone left after cutting off the top parallel to the base.

Chapter 14: Statistics

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Mean, median and mode of ungrouped data.
- Reading and constructing frequency tables.
- Ability to plot points and draw a smooth curve.
- Understanding of class intervals and class marks.

14.1 Mean of Grouped Data

 **Estimated time:** 15 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

The mean of grouped data is computed by the direct, assumed-mean or step-deviation method, all of which give the same answer.

Real-Life Example

A school computing the average marks of five hundred students uses grouped data because listing every individual score would be unmanageable.

Detailed Explanation

Definition

For grouped data, the mean is $\Sigma f_i x_i / \Sigma f_i$, where x_i is the class mark of the i th class and f_i its frequency. The step-deviation method uses $u_i = (x_i - a)/h$.

Key Idea

All three methods are algebraically identical; only the arithmetic burden differs. The step-deviation method is preferred when the class marks are large and the class size is uniform.

Working Principle

Grouping loses information: every observation in a class is treated as if it equalled the class mark. The computed mean is therefore an approximation, though a very good one.

Applications

- Direct method when the class marks are small.
- Assumed-mean method when the class marks are large but class sizes are unequal.
- Step-deviation method when class sizes are equal — it reduces the arithmetic most.

✓ **Points to Remember (Exam Focus)**

- ✓ Class mark $x_i = (\text{upper limit} + \text{lower limit})/2$.
- ✓ Direct method: $\text{mean} = \Sigma f_i x_i / \Sigma f_i$.
- ✓ Assumed mean: $\text{mean} = a + \Sigma f_i d_i / \Sigma f_i$, where $d_i = x_i - a$.
- ✓ Step-deviation: $\text{mean} = a + h(\Sigma f_i u_i / \Sigma f_i)$, where $u_i = (x_i - a)/h$.
- ✓ All three methods give exactly the same answer.

⚠ Common Student Mistakes

- ⚠ Using class limits instead of class marks as x_i .
- ⚠ Forgetting to multiply by h in the step-deviation method.
- ⚠ Choosing a value of a that is not a class mark, which complicates the arithmetic unnecessarily.



Memory Hook

'Class mark times frequency, divided by total frequency' — the other two methods are just shortcuts.



Advanced Insights

Grouping introduces a small bias when the distribution within classes is skewed. Sheppard's correction adjusts for this, though at school level the class-mark approximation is entirely adequate.

Formula Section



Mean of Grouped Data

Difficulty	● Medium ★★☆☆
Formula	Direct: $\bar{x} = \Sigma f_i x_i / \Sigma f_i$; Step-deviation: $\bar{x} = a + h(\Sigma f_i u_i / \Sigma f_i)$
Meaning of Variables	x_i = class mark, f_i = frequency, a = assumed mean, h = class size, $u_i = (x_i - a)/h$.
When to Use	For any grouped frequency distribution; step-deviation when class sizes are equal.
Common Mistakes	Omitting the factor h in the step-deviation formula.
Shortcut / Memory Trick	Pick a as the class mark of the class with the highest frequency.

14.2 Median, Mode and Ogives

🕒 Estimated time: 16 mins Difficulty: ● Medium ★★☆☆ Exam Importance: ★★★★★

Quick Summary

The median and mode of grouped data are found from the median and modal classes using interpolation formulas, and the median can also be read from an ogive.

Real-Life Example

Median income is quoted rather than mean income precisely because a handful of very high earners would distort the average.

Detailed Explanation

Definition

The median class is the class in which the cumulative frequency first reaches or exceeds $n/2$. The modal class is the class with the highest frequency. An ogive is a cumulative frequency curve.

Key Idea

Both formulas interpolate within a class rather than assuming the answer sits at a class boundary. This is why the cumulative frequency of the preceding class appears in the median formula.

Working Principle

The point where the 'less than' and 'more than' ogives intersect gives the median directly, which is a useful graphical check on the computed value.

Applications

- Median is preferred for skewed data such as income or house prices.
- Mode is used when the most common value is what matters, such as shoe sizes to stock.
- Empirical relation: $3 \text{ Median} = \text{Mode} + 2 \text{ Mean}$.

✓ Points to Remember (Exam Focus)

- ✓ Median = $l + [(n/2 - cf)/f] \times h$.
- ✓ Mode = $l + [(f_1 - f_0)/(2f_1 - f_0 - f_2)] \times h$.
- ✓ Median class: the class where cumulative frequency first reaches $n/2$.
- ✓ The two ogives intersect at the median.
- ✓ Empirical relation: $3 \text{ Median} = \text{Mode} + 2 \text{ Mean}$.

Common Student Mistakes

-  Using the frequency instead of the cumulative frequency of the preceding class in the median formula.
-  Taking the modal class as the one with the largest class mark instead of the largest frequency.
-  Failing to convert discontinuous classes into continuous ones before applying the formulas.

Memory Hook

'Median needs cumulative frequency; Mode needs the neighbours of the tallest bar.'

Advanced Insights

The median minimises the sum of absolute deviations while the mean minimises the sum of squared deviations. Each average is the optimal answer to a different question about the data.

Formula Section

Median of Grouped Data

Difficulty	● Medium ★★☆☆
Formula	$\text{Median} = l + [(n/2 - cf)/f] \times h$
Meaning of Variables	l = lower limit of the median class, n = total frequency, cf = cumulative frequency of the preceding class, f = frequency of the median class, h = class size.
When to Use	For grouped data, especially when it is skewed.
Common Mistakes	Using the frequency of the median class as cf .
Shortcut / Memory Trick	Build the cumulative frequency column first, before anything else.

Mode of Grouped Data

Difficulty	● Medium ★★☆☆
Formula	$\text{Mode} = l + [(f_1 - f_0)/(2f_1 - f_0 - f_2)] \times h$
Meaning of Variables	l = lower limit of the modal class, f_1 = its frequency, f_0 = preceding frequency, f_2 = following frequency, h = class size.
When to Use	For grouped data where the most frequent value is required.
Common Mistakes	Mixing up f_0 and f_2 , which lie on either side of the modal class.
Shortcut / Memory Trick	The modal class is simply the row with the highest frequency.

II Quick Recap — Topics Covered So Far

- Mean of Grouped Data — The mean of grouped data is computed by the direct, assumed-mean or step-deviation method, all of which give the same answer.
- Median, Mode and Ogives — The median and mode of grouped data are found from the median and modal classes using interpolation formulas, and the median can also be read from an ogive.

Chapter 14 — End of Chapter Revision

All Memory Hooks

- Mean of Grouped Data: 'Class mark times frequency, divided by total frequency' — the other two methods are just shortcuts.
- Median, Mode and Ogives: 'Median needs cumulative frequency; Mode needs the neighbours of the tallest bar.'

All Common Mistakes

-  Using class limits instead of class marks as x_i .
-  Forgetting to multiply by h in the step-deviation method.
-  Choosing a value of a that is not a class mark, which complicates the arithmetic unnecessarily.
-  Using the frequency instead of the cumulative frequency of the preceding class in the median formula.
-  Taking the modal class as the one with the largest class mark instead of the largest frequency.
-  Failing to convert discontinuous classes into continuous ones before applying the formulas.

Formula Sheet

Formula	Meaning / Use
Mean of Grouped Data: Direct: $\bar{x} = \Sigma f_i x_i / \Sigma f_i$; Step-deviation: $\bar{x} = a + h(\Sigma f_i u_i / \Sigma f_i)$	For any grouped frequency distribution; step-deviation when class sizes are equal.
Median of Grouped Data: Median = $l + [(n/2 - cf)/f] \times h$	For grouped data, especially when it is skewed.
Mode of Grouped Data: Mode = $l + [(f_1 - f_0)/(2f_1 - f_0 - f_2)] \times h$	For grouped data where the most frequent value is required.

Quick Revision Section

- Mean of Grouped Data: The mean of grouped data is computed by the direct, assumed-mean or step-deviation method, all of which give the same answer.
- Median, Mode and Ogives: The median and mode of grouped data are found from the median and modal classes using interpolation formulas, and the median can also be read from an ogive.

Chapter 15: Probability

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Empirical probability from Class 9.
- Ability to count outcomes systematically.
- Familiarity with a deck of cards, dice and coins.
- Comfort with fractions in lowest terms.

15.1 Theoretical Probability

 **Estimated time:** 16 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

Theoretical probability of an event is the number of favourable outcomes divided by the total number of equally likely outcomes.

Real-Life Example

The probability of drawing an ace from a well-shuffled deck is $4/52$ regardless of how many times you have tried before — the cards have no memory.

Detailed Explanation

Definition

For an experiment with equally likely outcomes, $P(E) = (\text{number of outcomes favourable to } E) / (\text{total number of possible outcomes})$. Two events are complementary if exactly one of them must occur.

Key Idea

The phrase 'equally likely' is essential. A biased die or a bag with differently sized balls breaks the assumption, and the formula no longer applies.

Working Principle

For two dice there are 36 equally likely outcomes, not 12 — the two dice are distinguishable even if identical in appearance. Listing the sample space as ordered pairs avoids this classic error.

Applications

- A deck has 52 cards: 26 red, 26 black, 4 suits of 13, 12 face cards.
- Two coins give 4 outcomes: HH, HT, TH, TT — not 3.
- $P(E) + P(\text{not } E) = 1$, so the complement is often the quicker route.

Points to Remember (Exam Focus)

- ✓ $P(E)$ = favourable outcomes / total outcomes, for equally likely outcomes.
- ✓ $0 \leq P(E) \leq 1$; $P(\text{impossible}) = 0$ and $P(\text{certain}) = 1$.
- ✓ $P(\text{not } E) = 1 - P(E)$.
- ✓ Two dice give 36 outcomes; two coins give 4.
- ✓ A deck of 52 cards has 4 aces, 12 face cards and 26 cards of each colour.

⚠ Common Student Mistakes

- ⚠ Counting the outcomes of two dice as 12 or 21 instead of 36.
- ⚠ Assuming outcomes are equally likely when they are not.
- ⚠ Treating the ace as a face card; the face cards are jack, queen and king only.



Memory Hook

'Favourable over Total' — and when the favourable cases are messy, count the complement instead.



Advanced Insights

Equally likely outcomes are an assumption, not a fact, and a badly chosen sample space gives wrong answers. Bertrand's paradox shows that the same physical question can yield different probabilities depending on how outcomes are parameterised.

Formula Section



Theoretical Probability

Difficulty	● Easy ★☆☆
Formula	$P(E) = n(E) / n(S)$
Meaning of Variables	$n(E)$ = number of favourable outcomes, $n(S)$ = total number of equally likely outcomes.
When to Use	For dice, coins, cards and any experiment with symmetric outcomes.
Common Mistakes	Miscounting the sample space, especially for two dice or two coins.
Shortcut / Memory Trick	Use $P(\text{not } E) = 1 - P(E)$ whenever 'at least one' appears in the question.

II Quick Recap — Topics Covered So Far

- Theoretical Probability — Theoretical probability of an event is the number of favourable outcomes divided by the total number of equally likely outcomes.

Chapter 15 — End of Chapter Revision

All Memory Hooks

- Theoretical Probability: 'Favourable over Total' — and when the favourable cases are messy, count the complement instead.

All Common Mistakes

-  Counting the outcomes of two dice as 12 or 21 instead of 36.
-  Assuming outcomes are equally likely when they are not.
-  Treating the ace as a face card; the face cards are jack, queen and king only.

Formula Sheet

Formula	Meaning / Use
Theoretical Probability: $P(E) = n(E) / n(S)$	For dice, coins, cards and any experiment with symmetric outcomes.

Quick Revision Section

- Theoretical Probability: Theoretical probability of an event is the number of favourable outcomes divided by the total number of equally likely outcomes.

Chapter 16: Binomial Theorem

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Factorial notation and combinations nC_r .
- Laws of exponents.
- Algebraic identities such as $(a + b)^2$ and $(a + b)^3$.
- Ability to simplify expressions with powers.

16.1 Binomial Expansion and General Term

Estimated time: 16 mins **Difficulty:** Hard **★★★** **Exam Importance:**

Quick Summary

The binomial theorem expands $(a + b)^n$ as a sum of $n + 1$ terms whose coefficients are the combinations nC_r .

Real-Life Example

Compound interest, probability distributions and approximations in physics all rely on binomial expansions rather than repeated multiplication.

Detailed Explanation

Definition

For any positive integer n , $(a + b)^n = \sum {}^nC_r a^{n-r} b^r$ for $r = 0$ to n . The general term is $T_{r+1} = {}^nC_r a^{n-r} b^r$.

Key Idea

The general term is indexed T_{r+1} , not T_r , because r starts at 0. This off-by-one is the single most common error in the whole chapter.

Working Principle

The powers of a decrease from n to 0 while the powers of b increase from 0 to n , and in every term the two powers add to n — a useful check on any expansion.

Applications

- The expansion has exactly $n + 1$ terms.
- The coefficients are the entries of the n th row of Pascal's triangle.
- Coefficients equidistant from the two ends are equal, since ${}^nC_r = {}^nC_{n-r}$.

Points to Remember (Exam Focus)

- ✓ $(a + b)^n = {}^nC_0 a^n + {}^nC_1 a^{n-1} b + \dots + {}^nC_n b^n$.

- ✓ General term $T_{r+1} = {}^nC_r a^{n-r} b^r$.
- ✓ Number of terms in the expansion = $n + 1$.
- ✓ The sum of the exponents in each term is always n .
- ✓ For $(a - b)^n$ the signs alternate, starting positive.

⚠ Common Student Mistakes

- ⚠ Writing the general term as T_r instead of T_{r+1} .
- ⚠ Forgetting the alternating signs in $(a - b)^n$.
- ⚠ Using n instead of $n + 1$ for the number of terms.

🧠 Memory Hook

'Powers of a fall, powers of b rise, and together they always add to n.'

🚀 Advanced Insights

Newton extended the theorem to negative and fractional exponents, where the expansion becomes an infinite series valid for $|x| < 1$. That generalisation is the basis of most approximation formulas in physics.

Formula Section

📐 Binomial Theorem

Difficulty	● Hard ★★★
Formula	$(a + b)^n = \sum_{r=0}^n {}^nC_r a^{n-r} b^r$
Meaning of Variables	a and b = the two terms, n = positive integer exponent, r = term index.
When to Use	To expand a binomial power or to find one particular term without expanding fully.
Common Mistakes	Interchanging the powers of a and b .
Shortcut / Memory Trick	For the middle term of an even power n , take $r = n/2$.

📐 General Term

Difficulty	● Medium ★★☆☆
Formula	$T_{r+1} = {}^nC_r a^{n-r} b^r$
Meaning of Variables	$r = 0, 1, 2, \dots, n$; T_{r+1} is the $(r+1)$ th term.

When to Use	To find a specific term, the term independent of x , or the coefficient of a given power.
Common Mistakes	Setting r equal to the required term number instead of one less.
Shortcut / Memory Trick	For the term independent of x , set the total power of x to zero and solve for r .

II Quick Recap — Topics Covered So Far

- Binomial Expansion and General Term — The binomial theorem expands $(a + b)^n$ as a sum of $n + 1$ terms whose coefficients are the combinations nC_r .

Chapter 16 — End of Chapter Revision

All Memory Hooks

- Binomial Expansion and General Term: 'Powers of a fall, powers of b rise, and together they always add to n.'

All Common Mistakes

-  Writing the general term as T_r instead of T_{r+1} .
-  Forgetting the alternating signs in $(a - b)^n$.
-  Using n instead of $n + 1$ for the number of terms.

Formula Sheet

Formula	Meaning / Use
Binomial Theorem: $(a + b)^n = \sum_{r=0}^n {}^nC_r a^{n-r} b^r$	To expand a binomial power or to find one particular term without expanding fully.
General Term: $T_{r+1} = {}^nC_r a^{n-r} b^r$	To find a specific term, the term independent of x , or the coefficient of a given power.

Quick Revision Section

- Binomial Expansion and General Term: The binomial theorem expands $(a + b)^n$ as a sum of $n + 1$ terms whose coefficients are the combinations nC_r .

Chapter 17: Sets, Relations and Functions

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Familiarity with the idea of a collection of objects.
- Understanding of ordered pairs and the Cartesian plane.
- Ability to substitute into an algebraic expression.
- Knowledge of number systems N , Z , Q and R .

17.1 Sets and Set Operations

Estimated time: 15 mins **Difficulty:** Medium **★★☆** **Exam Importance:**

Quick Summary

A set is a well-defined collection of distinct objects, and sets are combined using union, intersection, difference and complement.

Real-Life Example

A search filter showing items that are 'in stock AND under ₹500' is computing the intersection of two sets.

Detailed Explanation

Definition

A set is a well-defined collection of distinct objects. The union $A \cup B$ contains elements in A or B or both; the intersection $A \cap B$ contains elements in both; the difference $A - B$ contains elements in A but not in B .

Key Idea

'Well-defined' means membership must be decidable without opinion. 'The set of tall students' is not a set; 'the set of students above 170 cm' is.

Working Principle

Venn diagrams turn set identities into pictures, and the inclusion–exclusion principle $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ corrects for the double counting of the overlap.

Applications

- A set with n elements has 2^n subsets.
- De Morgan's laws: $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$.
- Disjoint sets have no common element, so $A \cap B = \emptyset$.

Points to Remember (Exam Focus)

- ✓ Elements of a set are distinct and unordered — $\{1,2,2,3\}$ is just $\{1,2,3\}$.
- ✓ The empty set $\emptyset$ is a subset of every set.
- ✓ A set with n elements has 2^n subsets and $2^n - 1$ proper subsets.
- ✓ $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.
- ✓ De Morgan's laws exchange union and intersection under complementation.

⚠ Common Student Mistakes

- ⚠ Listing repeated elements in a set.
- ⚠ Confusing $\in$ (element of) with $\subset$ (subset of).
- ⚠ Forgetting to subtract $n(A \cap B)$ and double counting the overlap.

🧠 Memory Hook

'Union is OR, Intersection is AND' — and subtract the overlap once to stop counting it twice.

🚀 Advanced Insights

Naive set theory permits Russell's paradox — the set of all sets that do not contain themselves. Modern mathematics avoids it with axiomatic set theory, which restricts how sets may be formed.

17.2 Relations and Functions

🕒 **Estimated time:** 15 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

🔑 Quick Summary

A relation is any set of ordered pairs, and a function is a relation in which each input has exactly one output.

🌐 Real-Life Example

A student's roll number maps to exactly one name, which makes it a function; a name might match several roll numbers, so the reverse is only a relation.

📖 Detailed Explanation

Definition

A relation R from set A to set B is a subset of the Cartesian product $A \times B$. A function $f : A \rightarrow B$ is a relation in which every element of A is associated with exactly one element of B .

Key Idea

Every function is a relation, but not every relation is a function. The distinguishing test is that no input may have two different outputs.

Working Principle

The vertical line test applies this graphically: if any vertical line meets the graph more than once, the relation is not a function.

Applications

- If $n(A) = m$ and $n(B) = n$, then $n(A \times B) = mn$ and there are 2^{mn} relations from A to B .
- Domain = set of all inputs; Range = set of actual outputs; Codomain = the target set B .
- One-one (injective), onto (surjective) and one-one onto (bijective) classify functions.

✓ Points to Remember (Exam Focus)

- ✓ A relation from A to B is a subset of $A \times B$.
- ✓ A function assigns exactly one output to each input.
- ✓ Domain, codomain and range must be distinguished; the range is a subset of the codomain.
- ✓ Vertical line test: more than one intersection means it is not a function.
- ✓ Number of relations from A to B is 2^{mn} where m and n are the sizes of A and B .

⚠ Common Student Mistakes

- ⚠ Calling every relation a function.
- ⚠ Using range and codomain interchangeably.
- ⚠ Assuming a function cannot give the same output for two different inputs — it can, unless it is one-one.



Memory Hook

'One input, one output' — that single rule separates functions from relations.



Advanced Insights

Defining a function as a set of ordered pairs removes any appeal to a 'rule' or 'formula', which matters because most functions cannot be written as a formula at all. This set-theoretic definition is what modern analysis is built on.

II Quick Recap — Topics Covered So Far

- Sets and Set Operations — A set is a well-defined collection of distinct objects, and sets are combined using union, intersection, difference and complement.

- Relations and Functions — A relation is any set of ordered pairs, and a function is a relation in which each input has exactly one output.

Chapter 17 — End of Chapter Revision

All Memory Hooks

- Sets and Set Operations: 'Union is OR, Intersection is AND' — and subtract the overlap once to stop counting it twice.
- Relations and Functions: 'One input, one output' — that single rule separates functions from relations.

All Common Mistakes

-  Listing repeated elements in a set.
-  Confusing $\in$ (element of) with $\subset$ (subset of).
-  Forgetting to subtract $n(A \cap B)$ and double counting the overlap.
-  Calling every relation a function.
-  Using range and codomain interchangeably.
-  Assuming a function cannot give the same output for two different inputs — it can, unless it is one-one.

Quick Revision Section

- Sets and Set Operations: A set is a well-defined collection of distinct objects, and sets are combined using union, intersection, difference and complement.
- Relations and Functions: A relation is any set of ordered pairs, and a function is a relation in which each input has exactly one output.

Chapter 18: Complex Numbers

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Quadratic equations and the discriminant.
- Operations with surds.
- Plotting points in the Cartesian plane.
- Pythagoras theorem.

18.1 Complex Numbers and Their Algebra

 **Estimated time:** 16 mins **Difficulty:**  Hard **★★★** **Exam Importance:**     

Quick Summary

A complex number has the form $a + ib$ where $i = \sqrt{-1}$, and complex numbers are added, multiplied and divided using ordinary algebra together with $i^2 = -1$.

Real-Life Example

Alternating-current circuits are analysed with complex impedance, because it captures both the size and the timing of a signal in one number.

Detailed Explanation

Definition

A complex number is of the form $z = a + ib$, where a is the real part, b is the imaginary part and $i = \sqrt{-1}$ with $i^2 = -1$. The conjugate of z is $\bar{z} = a - ib$ and the modulus is $|z| = \sqrt{a^2 + b^2}$.

Key Idea

Complex numbers exist so that every quadratic has roots. Once i is admitted, the Fundamental Theorem of Algebra guarantees that every polynomial of degree n has exactly n roots.

Working Principle

Division is carried out by multiplying numerator and denominator by the conjugate of the denominator, which makes the denominator real because $(a + ib)(a - ib) = a^2 + b^2$.

Applications

- Powers of i cycle with period 4: $i, -1, -i, 1$.
- Roots of a quadratic with negative discriminant are complex conjugates.
- The Argand plane represents $a + ib$ as the point (a, b) .

✓ **Points to Remember (Exam Focus)**

- ✓ $i = \sqrt{-1}$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$; powers repeat with period 4.
- ✓ $z = a + ib$; conjugate $\bar{z} = a - ib$; modulus $|z| = \sqrt{a^2 + b^2}$.
- ✓ $z \cdot \bar{z} = a^2 + b^2 = |z|^2$, which is always real and non-negative.
- ✓ Divide by multiplying numerator and denominator by the conjugate.
- ✓ Complex roots of a real polynomial always occur in conjugate pairs.

⚠ Common Student Mistakes

- ⚠ Writing $\sqrt{(-4)} \times \sqrt{(-9)} = \sqrt{36} = 6$; the correct value is $6i^2 = -6$.
- ⚠ Treating i as an ordinary variable and forgetting $i^2 = -1$.
- ⚠ Comparing two complex numbers as greater or less; they cannot be ordered.



Memory Hook

'i squared is minus one' — and the powers of i cycle every four.



Advanced Insights

Euler's formula $e^{i\theta} = \cos \theta + i \sin \theta$ unites exponentials with trigonometry, and at $\theta = \pi$ gives $e^{i\pi} + 1 = 0$, linking the five most important constants in mathematics in a single equation.

Formula Section



Modulus and Conjugate

Difficulty	● Medium ★★☆☆
Formula	$ z = \sqrt{a^2 + b^2}$; $\bar{z} = a - ib$; $z \cdot \bar{z} = z ^2$
Meaning of Variables	$z = a + ib$, a = real part, b = imaginary part.
When to Use	For division, rationalising and finding the distance of a point from the origin in the Argand plane.
Common Mistakes	Forgetting to change the sign of only the imaginary part when conjugating.
Shortcut / Memory Trick	$ z $ is just the distance formula applied to the point (a, b) .



Roots of a Quadratic with $D < 0$

Difficulty	● Medium ★★☆☆
Formula	$x = \frac{-b \pm i\sqrt{4ac - b^2}}{2a}$

Meaning of Variables	a, b, c = coefficients, with $b^2 - 4ac < 0$.
When to Use	When the discriminant is negative and complex roots are required.
Common Mistakes	Leaving a negative quantity under the root instead of extracting i .
Shortcut / Memory Trick	The two roots are always conjugates of each other.

II Quick Recap — Topics Covered So Far

- Complex Numbers and Their Algebra — A complex number has the form $a + ib$ where $i = \sqrt{-1}$, and complex numbers are added, multiplied and divided using ordinary algebra together with $i^2 = -1$.

Chapter 18 — End of Chapter Revision

All Memory Hooks

- Complex Numbers and Their Algebra: 'i squared is minus one' — and the powers of i cycle every four.

All Common Mistakes

-  Writing $\sqrt{(-4)} \times \sqrt{(-9)} = \sqrt{36} = 6$; the correct value is $6i^2 = -6$.
-  Treating i as an ordinary variable and forgetting $i^2 = -1$.
-  Comparing two complex numbers as greater or less; they cannot be ordered.

Formula Sheet

Formula	Meaning / Use
Modulus and Conjugate: $ z = \sqrt{a^2 + b^2}$; $\bar{z} = a - ib$; $z \cdot \bar{z} = z ^2$	For division, rationalising and finding the distance of a point from the origin in the Argand plane.
Roots of a Quadratic with $D < 0$: $x = [-b \pm i\sqrt{4ac - b^2}] / 2a$	When the discriminant is negative and complex roots are required.

Quick Revision Section

- Complex Numbers and Their Algebra: A complex number has the form $a + ib$ where $i = \sqrt{-1}$, and complex numbers are added, multiplied and divided using ordinary algebra together with $i^2 = -1$.

Chapter 19: Matrices

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Comfort with arrays of numbers in rows and columns.
- Ability to solve simultaneous linear equations.
- Basic algebraic manipulation.
- Understanding of ordered pairs and indices.

19.1 Matrices and Their Operations

 **Estimated time:** 16 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

A matrix is a rectangular array of numbers; addition requires equal orders, and multiplication requires the columns of the first to match the rows of the second.

Real-Life Example

Every image on a screen is a matrix of pixel values, and rotating or brightening it is a matrix operation.

Detailed Explanation

Definition

A matrix of order $m \times n$ has m rows and n columns. Element a_{ij} lies in the i th row and j th column. Two matrices are equal if they have the same order and all corresponding elements are equal.

Key Idea

Matrix multiplication is not commutative: AB is generally not equal to BA , and one may be defined while the other is not. This is the single biggest departure from ordinary algebra.

Working Principle

For AB to be defined, the number of columns of A must equal the number of rows of B , and the product has order (rows of A) $\times$ (columns of B).

Applications

- Solving simultaneous equations by writing them as $AX = B$.
- Computer graphics uses matrices for rotation, scaling and translation.
- Transition matrices model how a system moves between states over time.

Points to Remember (Exam Focus)

- ✓ Order $m \times n$ means m rows and n columns — rows first, always.
- ✓ Addition and subtraction require matrices of exactly the same order.
- ✓ AB is defined only if columns of A = rows of B .
- ✓ $AB \neq BA$ in general; matrix multiplication is not commutative.
- ✓ Types: row, column, square, diagonal, scalar, identity, zero.

⚠ Common Student Mistakes

- ⚠ Multiplying matrices element by element instead of row by column.
- ⚠ Assuming $AB = BA$.
- ⚠ Writing the order as columns $\times$ rows.

🧠 Memory Hook

'Rows first, columns second' — and 'inner dimensions must match, outer dimensions give the answer'.

🚀 Advanced Insights

Matrix multiplication is non-commutative because it represents the composition of transformations, and rotating then reflecting is genuinely different from reflecting then rotating. The algebra is encoding geometry.

Formula Section

📐 Matrix Multiplication

Difficulty	● Medium ★★☆☆
Formula	$(AB)_{ij} = \sum_k a_{ik} \cdot b_{kj}$
Meaning of Variables	a_{ik} = element of A in row i , column k ; b_{kj} = element of B in row k , column j .
When to Use	Whenever two compatible matrices are multiplied.
Common Mistakes	Multiplying corresponding elements instead of taking row-by-column sums.
Shortcut / Memory Trick	$(m \times n)(n \times p)$ gives an $(m \times p)$ matrix; the inner n must match.

19.2 Determinants and the Inverse of a Matrix

🕒 Estimated time: 15 mins Difficulty: ● Hard ★★☆☆ Exam Importance: ★★★★★

Quick Summary

The determinant of a square matrix is a single number that decides whether the matrix has an inverse, and the inverse exists exactly when the determinant is non-zero.

Real-Life Example

A system of equations has a unique solution exactly when its coefficient determinant is non-zero — engineers check this before attempting to solve.

Detailed Explanation

Definition

For a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the determinant is $ad - bc$. A matrix is singular if its determinant is zero and non-singular otherwise. The inverse A^{-1} satisfies $AA^{-1} = A^{-1}A = I$.

Key Idea

The determinant is the gatekeeper. If it is zero the matrix has no inverse, and the corresponding system of equations has either no solution or infinitely many.

Working Principle

For a 2×2 matrix, the inverse is $(1/\det A) \times \text{adj } A$, where the adjoint is obtained by swapping the diagonal elements and changing the signs of the off-diagonal ones.

Applications

- $AX = B$ has the unique solution $X = A^{-1}B$ when $\det A \neq 0$.
- The determinant of a 2×2 matrix is the area of the parallelogram formed by its rows.
- $\det(AB) = \det(A) \times \det(B)$.

✓ Points to Remember (Exam Focus)

- ✓ $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$.
- ✓ A^{-1} exists if and only if $\det A \neq 0$.
- ✓ For a 2×2 matrix, $A^{-1} = (1/\det A) \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.
- ✓ $AA^{-1} = A^{-1}A = I$, the identity matrix.
- ✓ $\det(AB) = \det(A) \cdot \det(B)$ and $(AB)^{-1} = B^{-1}A^{-1}$.

⚠ Common Student Mistakes

- ⚠ Swapping the wrong pair of elements when forming the adjoint of a 2×2 matrix.
- ⚠ Attempting to invert a singular matrix.
- ⚠ Writing $(AB)^{-1} = A^{-1}B^{-1}$; the order must be reversed.

 Memory Hook

'Swap the diagonal, flip the signs off it, divide by the determinant.'

 Advanced Insights

Geometrically the determinant is the factor by which a transformation scales area. A zero determinant means the transformation collapses the plane onto a line, which is exactly why the information needed to invert it has been lost.

Formula Section **Inverse of a 2×2 Matrix**

Difficulty	● Hard ★★★
Formula	$A^{-1} = (1/(ad - bc)) \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$
Meaning of Variables	$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, with $ad - bc \neq 0$.
When to Use	To solve $AX = B$, or wherever a matrix inverse is required.
Common Mistakes	Negating the diagonal elements instead of the off-diagonal ones.
Shortcut / Memory Trick	Swap a and d, negate b and c, then divide by the determinant.

II Quick Recap — Topics Covered So Far

- **Matrices and Their Operations** — A matrix is a rectangular array of numbers; addition requires equal orders, and multiplication requires the columns of the first to match the rows of the second.
- **Determinants and the Inverse of a Matrix** — The determinant of a square matrix is a single number that decides whether the matrix has an inverse, and the inverse exists exactly when the determinant is non-zero.

Chapter 19 — End of Chapter Revision

All Memory Hooks

- Matrices and Their Operations: 'Rows first, columns second' — and 'inner dimensions must match, outer dimensions give the answer'.
- Determinants and the Inverse of a Matrix: 'Swap the diagonal, flip the signs off it, divide by the determinant.'

All Common Mistakes

-  Multiplying matrices element by element instead of row by column.
-  Assuming $AB = BA$.
-  Writing the order as columns $\times$ rows.
-  Swapping the wrong pair of elements when forming the adjoint of a 2×2 matrix.
-  Attempting to invert a singular matrix.
-  Writing $(AB)^{-1} = A^{-1}B^{-1}$; the order must be reversed.

Formula Sheet

Formula	Meaning / Use
Matrix Multiplication: $(AB)_{ij} = \sum_k a_{ik} \cdot b_{kj}$	Whenever two compatible matrices are multiplied.
Inverse of a 2×2 Matrix: $A^{-1} = (1/(ad - bc)) \begin{bmatrix} d, -b \\ -c, a \end{bmatrix}$	To solve $AX = B$, or wherever a matrix inverse is required.

Quick Revision Section

- Matrices and Their Operations: A matrix is a rectangular array of numbers; addition requires equal orders, and multiplication requires the columns of the first to match the rows of the second.
- Determinants and the Inverse of a Matrix: The determinant of a square matrix is a single number that decides whether the matrix has an inverse, and the inverse exists exactly when the determinant is non-zero.